Project Report On

Silent Speech Recognition for the Voiceless

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

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March 2026

CERTIFICATE

This is to certify that the project report entitled ***"Silent Speech Recognition for the Voiceless"*** *is a bonafide record of the work done by* ***Darsan P (U2203074), Fathima Meharin Irshad (U2203093), Geevar Saji Kuriakose (U2203096), Giribala Arun (U2203099)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Rev. Dr. Jaison Paul Mulerikkal**, Principal of RSET, and **Dr. Preetha K.G.**, Professor & Head of the Department of Computer Science and Engineering for providing us with the opportunity to undertake our project, "Silent Speech Recognition for the Voiceless".

We are highly indebted to our project coordinator, **Dr. Jincy J. Fernandez**, Associate Professor, Department of Computer Science and Engineering, for her valuable support.

It is also our pleasure and it gives us a moment of satisfaction for us to express our sincere gratitude to our project guide **Ms. Jisha Mary Jose**, Assistant Professor, Department of Computer Science and Engineering, for her patience and all the priceless advice and wisdom she has shared with us.

Last but not the least, we would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

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Abstract

Individuals suffering from speech impairments, like those affected by conditions such as Amyotrophic Lateral Sclerosis (ALS) or vocal cord paralysis, often face difficulties when communicating and rely on external assistive devices. Traditional Augmentative and Alternative Communication (AAC) devices may cause inconveniences due to their limited flexibility in working. To address these challenges, we propose a system that uses Deep Learning (DL) techniques to deliver a hassle-free approach that could convert silent lip movements into readable text and optionally into speech. The system employs a sentence-level model trained on the GRID dataset and a real-time inference approach. The former utilises Computer Vision (CV) techniques such as MediaPipe to detect facial landmarks. A strategic integration of Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (Bi-LSTM) facilitates processing the frames sequentially and interpreting them to produce readable text. This text can be converted to speech using a Text-to-Speech (TTS) engine. The latter approach focuses on the real-time deployment of the trained model using Flask, enabling a smooth communication experience. Unlike traditional AAC systems, this system requires minimal physical effort, making it suitable for individuals affected by speech and motor disabilities. In addition, a small custom dataset containing predefined emergency vocabulary recorded from multiple speakers was collected to fine-tune and evaluate the system under real-time conditions.

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List of Abbreviations

ALS – Amyotrophic Lateral Sclerosis

AAC – Augmentative and Alternative Communication

AI – Artificial Intelligence

DL – Deep Learning

CV – Computer Vision

GRID – GRammatical Image Dataset

CNN – Convolutional Neural Network

LSTM – Long Short-Term Memory

TTS – Text-to-Speech

VSR – Visual Speech Recognition

AVSR – Audio-Visual Speech Recognition

SNR – Signal-to-Noise Ratio

ALSOS – Amyotrophic Lateral Sclerosis Online System

CTC – Connectionist Temporal Classification

BGR – Blue-Green-Red

RGB – Red-Green-Blue

CER – Character Error Rate

UI – User Interface

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Chapter 1

Introduction

1.1 Background

Speech is one of the most natural and efficient means of human communication. Unfortunately, a significant portion of the population is affected by speech impairments resulting from medical conditions such as laryngectomy, neurological disorders, paralysis, or injury. Traditional assistive technologies like text-to-speech devices require manual input, making them time-consuming and unnatural for real-time communication. With advancements in AI and computer vision, it is now possible to use Visual Speech Recognition (VSR) or lip-reading systems (developed as Silent Speech Recognition Systems for the Voiceless) are designed to interpret silent speech by analyzing the visual movements of the lips, and facial regions. The proposed system builds on this concept to create an accessible tool for the voiceless community.

1.2 Problem Definition

Individuals who have lost their voice face extreme communication barriers in daily life. While sign language serves as one form of communication, it is not universally understood, leading to limitations. Existing speech-assistive devices depend heavily on residual vocal signals or manual text input, which restricts natural interaction. The challenge, therefore, lies in developing a non-verbal communication system that can interpret speech from visual input alone, without any audio cues. This project aims to address that challenge by designing a deep learning-based speech recognition system that takes lip movement as input and outputs meaningful text in real time.

1.3 Scope and Motivation

The scope of this project includes the design, implementation, and evaluation of a visual speech recognition system capable of recognizing spoken words or sentences purely from visual cues. The motivation behind this project stems from the need to assist individuals who cannot speak due to medical or congenital reasons, providing them with a reliable means of communication. Moreover, the system can find applications beyond medical use, such as in noisy environments where traditional voice recognition fails, security systems, military communications, or even voice-free human–computer interaction.
By combining the power of deep convolutional networks and recurrent neural architectures, this project aims to deliver a practical, lightweight, and affordable assistive tool that promotes inclusivity and accessibility.

1.4 Objectives

- To capture silent lip movements using a camera-enabled device.
- To detect and focus on the lip region for analysis.
- To interpret lip movements into accurate text in real time.
- To provide an easy-to-use, portable solution for everyday communication.

1.5 Risks and Challenges

The development of a visual speech recognition system comes with several challenges. One major issue is lighting and video quality, as inconsistent illumination can distort mouth detection accuracy. Variability in lip movements among different speakers, such as differences in pronunciation, speed, or facial features, makes model generalization difficult. Excessive head movements or occlusions, such as facial hair or accessories, can interfere with mouth tracking. Another challenge is the limited availability of large, annotated datasets, which affects model training and validation. Furthermore, the computational complexity of deep neural networks requires powerful hardware, which may not be feasible for all environments.

1.6 Assumptions

The system assumes that the user's face is well-lit and directly visible to the camera during recording. It is also assumed that the camera maintains a consistent frame rate to ensure smooth lip tracking and that users remain within a moderate distance from the lens. The model assumes prior training on a dataset that covers a wide range of speakers and that output text is to be generated in English.

1.7 Societal / Industrial Relevance

The societal relevance of this project lies in its potential to empower individuals who are voiceless or have severe speech impairments. By offering a natural and contactless means of communication, this system enhances quality of life and social inclusion. From an industrial perspective, the underlying technology can be adapted for applications in voice-free command systems, hands-free interfaces, silent communication tools, and assistive robotics. This aligns with the broader goal of building AI systems that are inclusive, accessible, and beneficial to diverse user groups.

1.8 Organization of the Report

This report is organized into four main chapters. The first chapter introduces the motivation, problem statement, and scope of the project. The second chapter provides a literature survey of existing research in the field of visual speech recognition and identifies the research gap. The third chapter describes the system design, architecture, algorithms, datasets, and implementation details. The final chapter concludes the report with observations, outcomes, and potential directions for future enhancement.

1.9 Summary of the Chapter

This introductory chapter outlined the background and motivation behind the project, discussed the problem definition, and presented the objectives and challenges involved in designing a visual speech recognition system. It also emphasized the societal importance of the proposed solution and laid the foundation for the subsequent chapters.

Chapter 2

Literature Survey

2.1 Introduction

The domain of visual speech recognition has seen rapid advancements in recent years due to progress in deep learning and computer vision. Early systems relied on handcrafted features and traditional machine learning methods, but modern approaches leverage convolutional and recurrent neural networks for end-to-end learning. This chapter reviews several key research works and identifies existing limitations that inspired this project.

2.2 Survey Papers

2.2.1 Audio-Visual Speech Recognition in Noisy Environments

In recent years, AVSR has significantly advanced through the adoption of deep learning techniques. Traditional audio-only Speech Recognition systems often struggle in noisy environments; however, AVSR systems leverage both auditory and visual modalities to enhance recognition accuracy. Deep neural networks, including convolutional and recurrent architectures, have been at the forefront of this progress. These systems effectively integrate spatial features from visual frames with temporal dynamics in audio, improving robustness in low signal-to-noise ratio (SNR) conditions. A key innovation in this domain involves the integration of attention mechanisms and sub-word-level modeling. Attention modules help AVSR models to selectively focus on relevant temporal or spatial segments, which is crucial when the visual input is ambiguous or partially occluded. Sub-word modeling enables finer-grained recognition capabilities, addressing co-articulation and other complex phonetic variations visible in lip movements [1][14].

The overall architecture of the proposed Audio-Visual Speech Recognition (AVSR) system is illustrated in Figure 2.1. The framework consists of four major components: an

audio speech recognition module, a visual speech recognition module, a cross-modal fusion mechanism, and a decoding module [2].

The audio frontend processes raw speech waveforms by dividing them into short frames. A pre-trained *wav2vec 2.0* model is used to extract high-level audio features from these frames. The extracted representations are typically high dimensional (1024 dimensions), so a 1D convolution layer is applied in the audio backend to compress the features to 512 dimensions. The visual frontend extracts lip movement patterns from the input video frames. Prior to feature extraction, RGB frames are converted into grayscale to reduce computational complexity. After extracting audio and visual representations, both modalities are combined through a cross-modal fusion mechanism. Let f_v denote the visual features and f_a denote the audio features. These features are concatenated to form a joint representation that captures complementary information from both modalities. A Seq2Seq Transformer decoder predicts the next character in the transcription sequence based on previous outputs. In parallel, a Connectionist Temporal Classification (CTC) decoder predicts character sequences without requiring explicit alignment between the audio input and text labels.

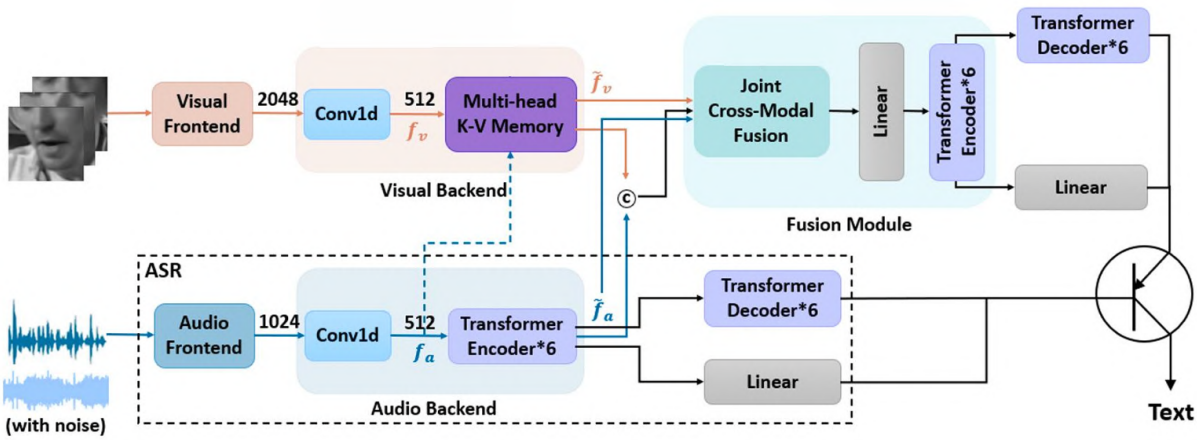


Figure 2.1: Overview of the Audio-Visual Speech Recognition (AVSR) model architecture, combining visual features from lip movements with temporal sequence modeling.

2.2.2 Lip Reading Using Alternating 2D-3D Convolutions

Recent approaches in lip reading focus on enhancing video-based feature extraction using convolutional architectures tailored for spatio-temporal data. One such innovation is the

ALSOS module, which alternates between 3D convolutions (capturing temporal motion across video frames) and 2D convolutions (capturing spatial details within frames). This alternating scheme enables more efficient and detailed feature learning than traditional convolutional pipelines. These ALSOS modules are integrated into deeper CNN architectures, providing multi-level spatio-temporal representations [2][10].

Figure 2.2 shows the workflow in the recent LR systems. Given an image sequence, the mouth regions are detected and cropped. Then, the frontend consists of a spatiotemporal convolution stage along with a CNN for capturing the short-term dynamics and finally extracting a feature vector for each frame [4][5]. The features are the inputs of the backend that is composed of the sequence modeling and classification stages. Thus, the information is encoded by a sequence modeling stage, and the output of the sequence modeling is being fed to the classification module, which provides the class probabilities, i.e., the decoded speech.

Figure 2.2: The workflow in the recent Lip Reading (LR) systems based on deep learning

2.2.3 Visual Speech Recognition Using LSTM Networks

Another prominent direction in VSR involves Recurrent Neural Networks (RNNs), particularly LSTM and Bi-LSTM networks. These architectures excel at capturing temporal dependencies, making them well-suited for modeling the sequence of visual features extracted from lip movements [1]. Typically, visual features are first extracted using 3D Convolutional Neural Networks (3D CNNs), which process spatial information across temporal dimensions. The resulting spatio-temporal embeddings are then fed into BiLSTMs, which capture both past and future contexts of each frame. To further enhance performance, attention mechanisms are integrated with LSTM-based models, allowing the system to dynamically focus on critical time steps or regions within the input sequence. This improves the model's ability to learn complex temporal structures [3][18].

Another important contribution in this area is the development of specialized datasets designed to support visual speech recognition research. One such dataset is the MobLip dataset, which was created to address the lack of publicly available VSR datasets in the Greek language. The MobLip dataset was collected as part of the Let's Talk! project, where thirty participants were asked to recite a unique one-minute text while being recorded on video. The recordings were captured at a sampling rate of 30 frames per second, ensuring sufficient temporal resolution for accurately analyzing lip movements and facial dynamics.

To prepare the dataset for model training, the annotated timestamps were converted into frame indices corresponding to the video frames as shown in Figure 2.3. This mapping allowed each spoken word to be associated with a sequence of lip movement frames. However, since different words naturally produce varying numbers of frames, oversampling and undersampling techniques were applied to maintain a consistent frame count for each word sequence during training [5]. Through this process, the dataset was curated into a large collection of approximately 55,000 images containing detailed lip movement information. The MobLip dataset thus provides a valuable resource for training and evaluating deep learning models for Greek-language visual speech recognition.

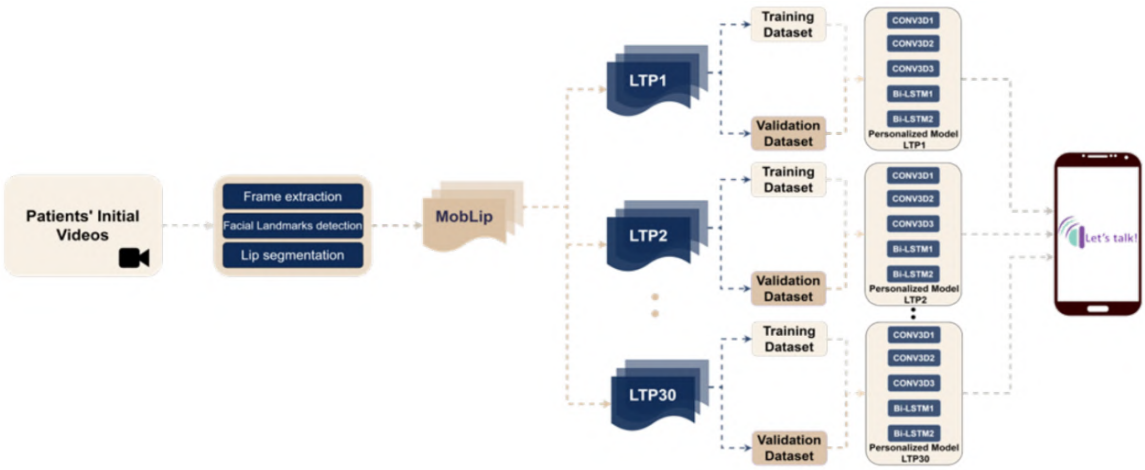


Figure 2.3: MobLip dataset creation, and the integration of personalized models into the mobile application

2.2.4 CTC-Based Visual Speech Recognition and End-to-End Models

Lipreading, also known as Visual Speech Recognition (VSR), involves interpreting speech by analyzing lip, tongue, and facial movements. While effective in controlled settings, these methods struggled under real-world conditions with variations in lighting, head pose, and background. The adoption of deep learning, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), enabled automatic feature extraction from raw video data, reducing dependence on handcrafted inputs. More recent approaches incorporate Transformers and hybrid CTC/Attention models, allowing end-to-end training from video sequences to text output and significantly improving performance in unconstrained scenarios [4][14].

Preprocessing is a critical step in lipreading systems to isolate the most informative visual features. Face detection and alignment models such as MTCNN or RetinaFace ensure normalized facial inputs, while the mouth region is cropped as the Region of Interest (ROI) to focus on lip movements. Frames are then resized, converted to grayscale, normalized, and temporally aligned using sampling or padding. Data augmentation strategies, including random cropping, flipping, or noise injection, further improve model robustness. Lipreading is fundamentally a sequence-to-sequence problem, where a variable-length frame sequence must be mapped to text. Early sequence modeling relied on HMMs and later LSTMs or GRUs, but these methods struggled with variable-length alignment between input frames and linguistic units. The introduction of Connectionist Temporal Classification (CTC) addressed these challenges by enabling alignment-free learning from unsegmented data, while hybrid CTC-Attention models leverage the monotonic alignment of CTC along with the contextual flexibility of attention mechanisms [15].

Figure 2.4 illustrates a typical sequence modeling architecture in VSR, combining CNN-based feature extraction, recurrent layers for temporal modeling, and a CTC transcription layer.

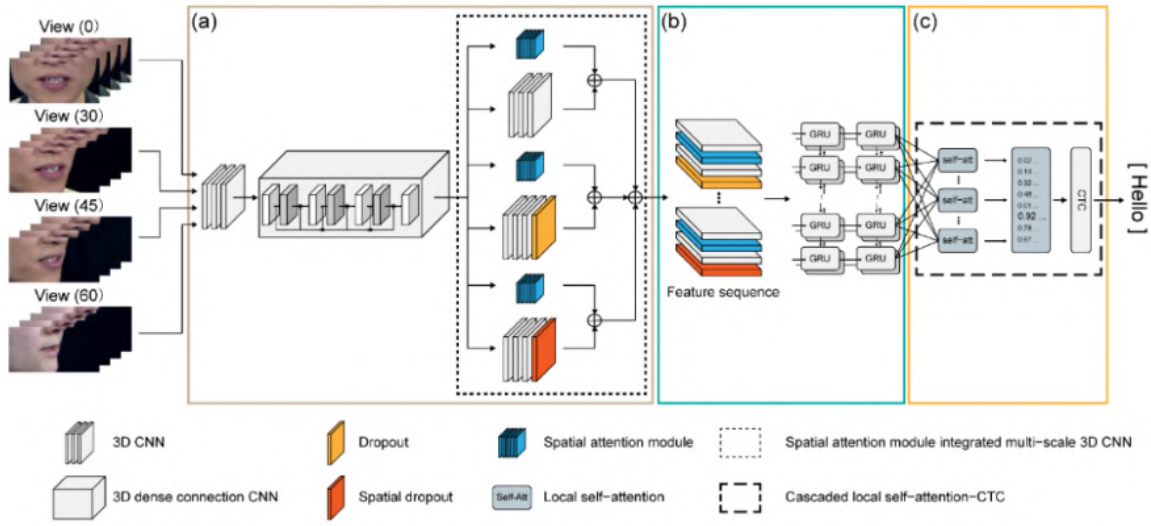


Figure 2.4: Sequence modeling in lipreading with CNN encoder, recurrent layers, and CTC transcription.

2.3 Summary and Gaps Identified

2.3.1 Comparison of the Papers

The survey of recent literature highlights the significant progress made in Audio-Visual Speech Recognition (AVSR) and Visual Speech Recognition (VSR) through the adoption of deep learning architectures. Modern AVSR systems combine audio and visual modalities to improve robustness in challenging conditions such as noisy environments. Architectures incorporating pre-trained audio models such as wav2vec 2.0, convolutional neural networks for visual feature extraction, and Transformer-based sequence modeling have demonstrated strong performance in multimodal speech recognition tasks. The integration of attention mechanisms and sub-word level modeling further enhances recognition accuracy by enabling models to focus on relevant temporal and spatial features.

Similarly, advancements in lip reading have focused on improving spatio-temporal feature extraction from video sequences. Techniques such as alternating 2D–3D convolutional modules enable efficient capture of both spatial information within frames and temporal motion across frames. Additionally, recurrent architectures such as LSTM and Bi-LSTM networks have been widely used for modeling temporal dependencies in visual speech data. End-to-end frameworks using CTC-based decoding have also gained popularity, allowing

models to perform sentence-level lipreading without requiring precise alignment between input frames and output text.

Table 2.1: Summary of Related Work in Visual Speech Recognition

Paper	**Main Focus**	**Model Type**	**Applications**
Audio-Visual Speech Recognition in Noisy Environments	Combining audio and visual inputs for speech recognition	CNNs, LSTMs, AVSR hybrid	Robust to noisy environments and utilizes both auditory and visual cues
Lip Reading Using Alternating 2D-3D Convolutions	Lipreading system using deep learning and convolutional models	ALSOS + ResNet CNN	Improved recognition performance through spatiotemporal feature extraction
Visual Speech Recognition Using LSTM Networks	Deep learning models for lipreading with temporal modeling	LSTM-based VSR	Strong handling of temporal dependencies in lip movement sequences
Lipreading with CTC-Based Visual Speech Recognition	Sentence-level Visual Speech Recognition (VSR)	CTC-based VSR model	Direct video-to-text mapping without requiring frame-level alignment

2.3.2 Gaps Identified

Despite these advancements, several research gaps remain in the field. First, many existing models rely heavily on large-scale datasets and computationally expensive architectures, which may limit their deployment in real-time or resource-constrained environments. Second, the availability of high-quality annotated datasets remains limited, particularly for low-resource languages. While datasets such as MobLip have attempted to address this challenge for the Greek language, similar datasets are still lacking for many other languages and dialects [1][3].

Another limitation lies in the robustness of visual-only systems under real-world condi-

tions. Factors such as lighting variations, occlusions, head movements, and differences in speaking styles can significantly impact recognition accuracy. Additionally, although multimodal fusion techniques improve performance, effectively learning complementary information from audio and visual streams remains an open research challenge.

Therefore, further research is needed to develop more efficient and generalizable AVSR and VSR models, improve cross-modal fusion strategies, and expand the availability of diverse annotated datasets. Addressing these gaps will contribute to building more robust speech recognition systems capable of operating effectively in real-world environments.

2.4 Summary of the Chapter

This chapter presented an overview of major research efforts in visual speech recognition and identified the challenges and opportunities in the field. The insights gained from the literature survey guided the architectural and algorithmic decisions for the proposed system described in the next chapter.

Chapter 3

System Design

3.1 Overall System Architecture

The overall system architecture of the proposed lip reading system is designed as a modular pipeline that integrates video preprocessing, deep learning-based feature extraction, sequence modeling, and real-time output generation [1][9].

As shown in Fig. 3.1, the architecture for the GRID dataset follows a sentence-level prediction pipeline. The system begins with video input acquired either from pre-recorded dataset samples or live video streams. The preprocessing module extracts frames, detects facial landmarks using MediaPipe FaceMesh, and isolates the lip region of interest (ROI). The cropped frames are then resized, converted from BGR to RGB format, and normalized to ensure consistency with model input requirements [8][13].

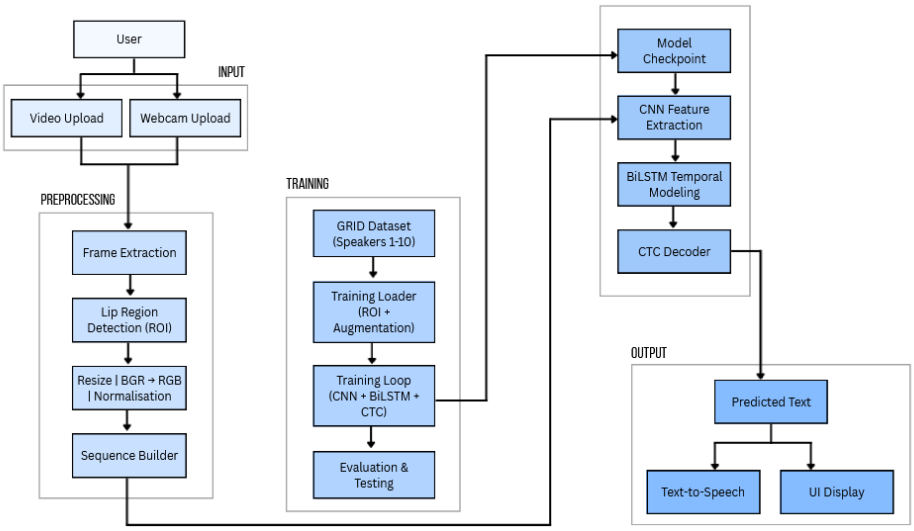


Figure 3.1: System Architecture - GRID

The processed frame sequences are passed to a Convolutional Neural Network (CNN), specifically a pre-trained ResNet-18, which extracts spatial features from each frame.

These features are then fed into a Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal dependencies across the sequence. For sentence-level prediction, a Connectionist Temporal Classification (CTC) decoder is used to generate text output without requiring explicit frame-level alignment [7][12].

To enhance real-time performance, a fine-tuned architecture is introduced using a custom dataset, as illustrated in Fig. 3.2. While the preprocessing and feature extraction stages remain the same, the model is adapted for word-level prediction. The fine-tuned model replaces the CTC decoding mechanism with a Softmax-based classifier, enabling faster and more efficient prediction for a limited vocabulary [1].

During inference, the system processes incoming video frames through the trained model and generates predictions in real time. The output text is displayed via a user interface and can optionally be converted into speech using a Text-to-Speech module, providing an assistive communication solution [13].

Overall, the architecture effectively combines robust preprocessing, deep learning-based spatial and temporal modeling, and flexible output mechanisms. The integration of both GRID-based sentence-level modeling and custom dataset fine-tuning allows the system to achieve a balance between generalization and real-time efficiency.

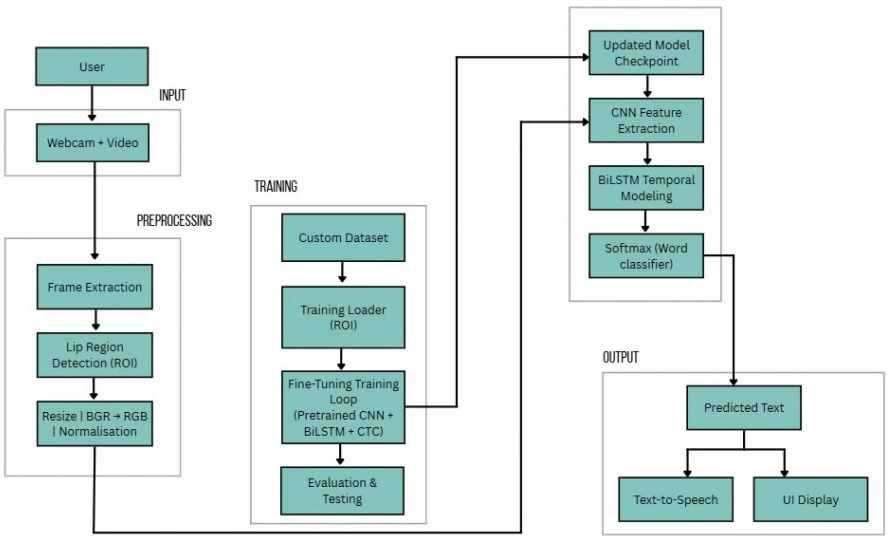


Figure 3.2: System Architecture - Custom

3.2 Design Diagrams

The sequence diagrams illustrate the dynamic interaction between different components of the proposed system during execution for both the GRID dataset and the custom dataset pipelines.

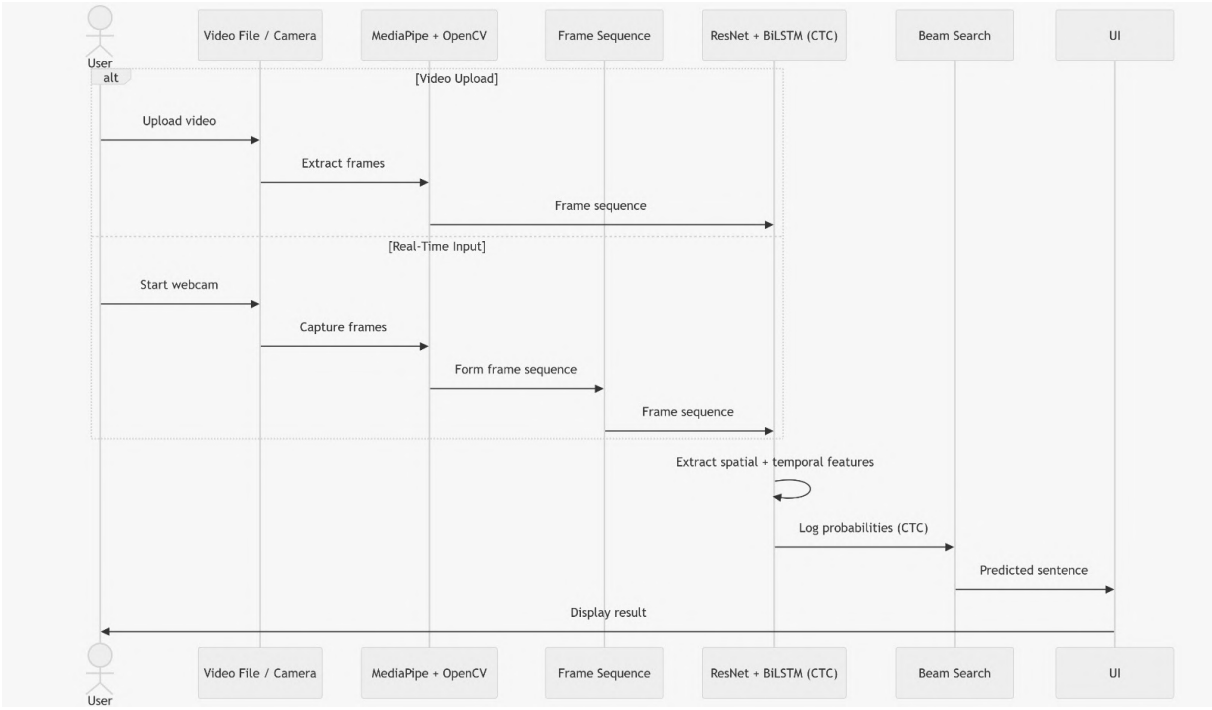


Figure 3.3: Sequence Diagram - GRID

As shown in Fig. 3.3, the sequence diagram for the GRID-based system represents the sentence-level lip reading workflow. The process begins with the user providing input through a video stream, which is then processed by the preprocessing module. Frames are extracted, and the lip region of interest (ROI) is detected and normalized. The processed frame sequence is passed through the CNN for spatial feature extraction, followed by the BiLSTM network for temporal modeling. The model generates probability distributions over character sequences, which are decoded using a Connectionist Temporal Classification (CTC) decoder to produce the final predicted sentence. The output is then displayed through the user interface and optionally converted to speech.

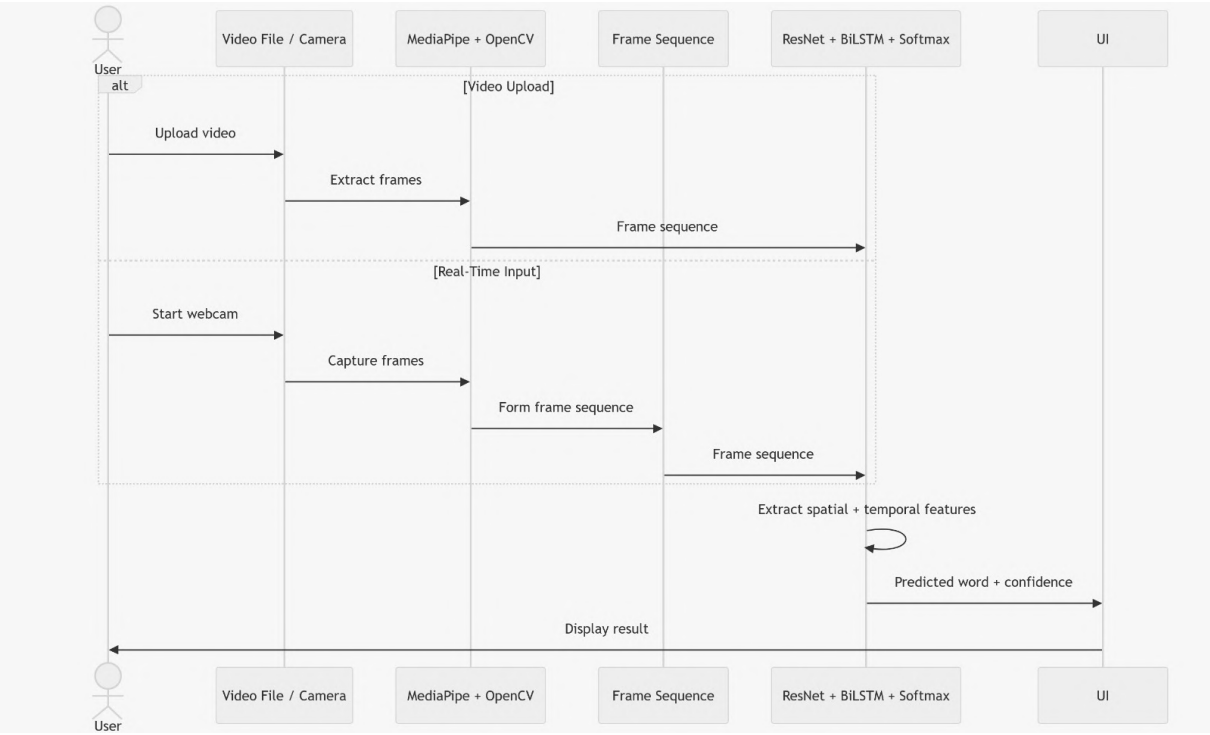


Figure 3.4: Sequence Diagram - Custom

Similarly, Fig. 3.4 illustrates the sequence of operations for the custom dataset pipeline, which is optimized for real-time word-level prediction. The initial preprocessing and feature extraction stages remain consistent with the GRID pipeline. However, instead of using a CTC decoder, the fine-tuned model employs a Softmax-based classifier to directly predict words from a predefined vocabulary. This modification reduces inference latency and improves efficiency for real-time applications. The predicted output is then displayed to the user and can also be passed to a Text-to-Speech module for audio feedback.
These sequence diagrams highlight the flow of data and control across system components, clearly demonstrating the transition from raw video input to meaningful textual output in both general-purpose and application-specific scenarios.

3.3 Module Design

3.3.1 Data Acquisition

GRID Model

The system processes inputs from archival video files as well as live video streams (Figure 3.5). To reduce the computational over- head of processing full-frame videos, a pre-

processing pipeline is applied.

Video sequences are split into temporal frames for analysis. The MediaPipe FaceMesh detector identifies 468 facial landmarks, from which lip boundary points (e.g., 61, 291, 0, 17, 13, and 14) are used to compute a bounding box around the perioral region. This allows dynamic extraction of the Region of Interest (ROI) corresponding to the mouth area. Video frames are read using OpenCV, which loads them in BGR format, while MediaPipe expects RGB input [6]. Therefore, a BGR-to-RGB conversion is performed prior to landmark detection. The cropped lip frames are then resized to a fixed spatial resolution to match the neural network input. During model training and inference, frames are converted to tensor representations and normalized using

$$T = \frac{x - 0.5}{0.5}$$

which scales pixel values from $[0, 1]$ to $[-1, 1]$. Finally, the processed frames are stacked into ordered temporal sequences to preserve the dynamics of lip motion [8][9].

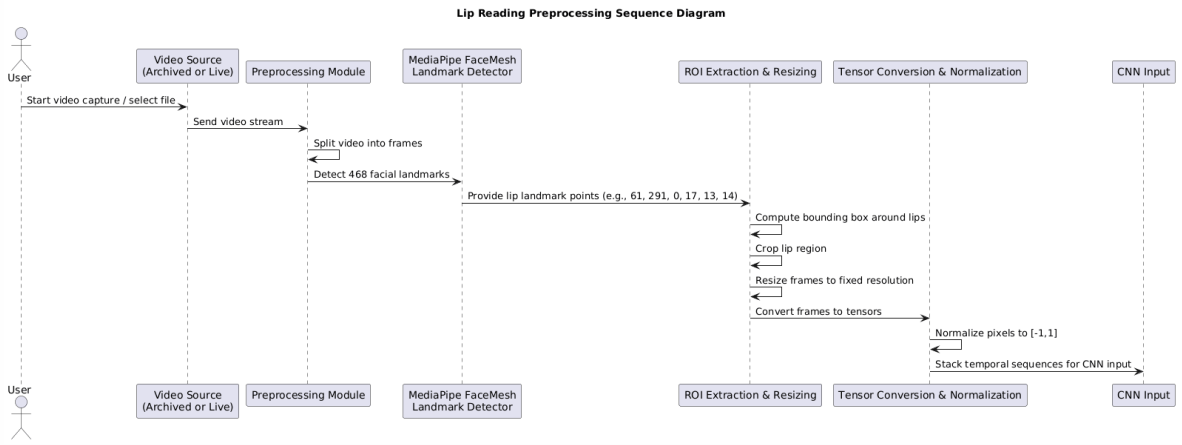


Figure 3.5: GRID Module 1

Custom Model

The custom dataset was developed to enable word-level lip reading tailored for real-time applications. Lip movement videos were recorded using webcams and smartphone cameras under different lighting conditions and backgrounds to improve robustness (Figure 3.6). Multiple speakers were included to capture variations in lip shape, speaking speed, and articulation patterns. Each recording corresponds to a predefined word from a limited

vocabulary set (help, water, yes, no, pain).

The recorded videos were processed frame-by-frame using OpenCV. MediaPipe FaceMesh was employed to detect facial landmarks, from which specific lip landmark points were used to extract the region of interest (ROI) corresponding to the mouth area. The extracted lip regions were resized to a fixed resolution (64×64 pixels) and saved as image sequences. The dataset was organized into structured directories based on speaker, word label, and train/test splits, and corresponding CSV manifest files were generated to map frame sequences to their respective labels.

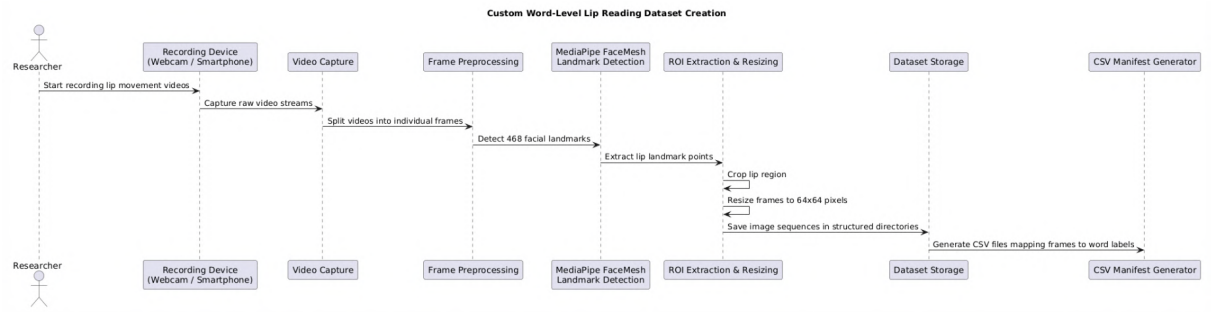


Figure 3.6: Custom Module 1

3.3.2 Feature Extraction and Model Training

GRID Model

The proposed system employs a hybrid CNN–BiLSTM–CTC architecture for sentence-level visual speech recognition, trained primarily on the GRID dataset. A pre-trained ResNet-18 backbone is used as a frame-level feature extractor. The final classification layer of ResNet-18 is removed, and the convolutional layers are frozen during training. This leverages pre-trained visual features learned on ImageNet while reducing the number of trainable parameters, thereby improving training efficiency and generalization [7][12]. The frame-level features are then sequentially fed into a two-layer Bidirectional Long Short-Term Memory (Bi-LSTM) network with a hidden size of 256. The Bi-LSTM processes the feature sequence in both forward and backward directions [8], allowing the network to capture contextual dependencies from both past and future frames. This bidirectional temporal modeling is crucial in lipreading, as the articulation of a phoneme or word can be influenced by surrounding frames.

To handle the alignment problem (Figure 3.7) between input frame sequences and output text sequences of variable length, the model employs Connectionist Temporal Classification (CTC). The CTC loss function is defined as:

$$\mathcal{L}_{CTC} = -\ln p(y \mid x), \tag{3.1}$$

where the probability of a target label sequence y given input sequence x is:

$$p(y \mid x) = \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t=1}^{T} p(\pi_t \mid x). \tag{3.2}$$

Here, π represents all valid CTC paths, $\mathcal{B}$ is the collapse function that removes repeated labels and blanks, and T is the sequence length in time steps. By summing over all possible alignments, CTC enables the model to learn mappings from video frames to text without requiring explicit frame-level labels. During training, the GRID dataset is divided into training and validation subsets. Model performance is measured using Character Error Rate (CER), defined as:

$$CER = \frac{S + D + I}{N}, \tag{3.3}$$

where S, D, and I denote the number of substitutions, deletions, and insertions, respectively, and N is the total number of characters in the reference transcript. The checkpoint with the lowest validation CER is saved for inference.

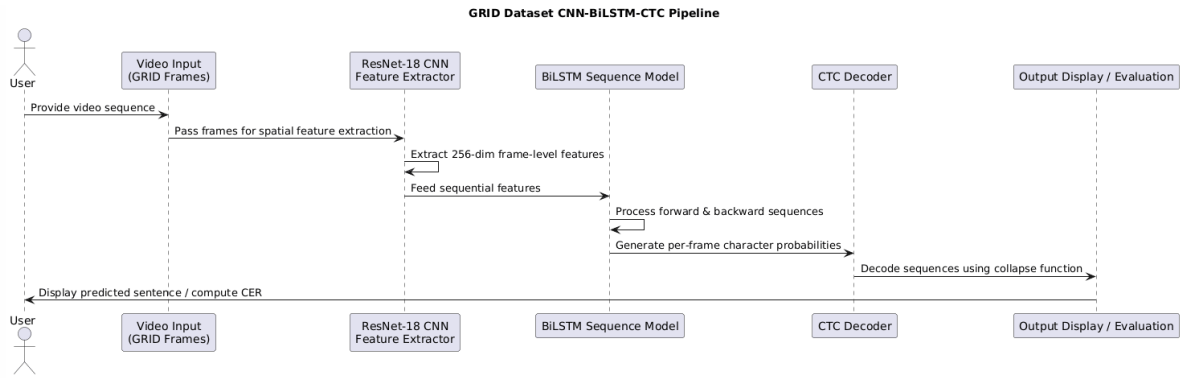


Figure 3.7: GRID Module 2

Custom Model

The preprocessed lip frame sequences were used as input for training the deep learning model (Figure 3.8). Each sequence was standardized to a fixed length (75 frames)

to ensure consistent temporal input. Spatial feature extraction was performed using a Convolutional Neural Network (CNN) based on the ResNet-18 architecture, which was pretrained on large-scale image data to capture robust visual features such as lip contours, motion patterns, and texture.

The extracted frame-level features were then passed to a Bidirectional Long Short-Term Memory (BiLSTM) network, which models temporal dependencies by processing the sequence in both forward and backward directions. This enables the model to understand the dynamic progression of lip movements over time.

Transfer learning was applied by initializing the model with weights trained on the GRID dataset, allowing reuse of previously learned spatio-temporal features. During fine-tuning, selected layers of the network were updated while preserving general feature representations. Unlike the GRID setup, which uses CTC decoding for sequence prediction, the custom model employs a softmax-based word-level classifier to predict one word from a predefined vocabulary. The model was trained using backpropagation and optimized using the AdamW optimizer, with cross-entropy loss guiding the classification task.

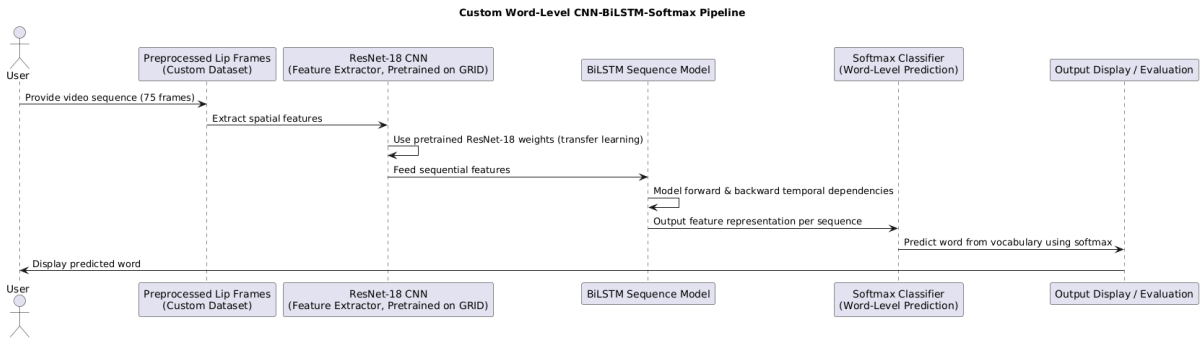


Figure 3.8: Custom Module 2

3.3.3 Inference and Output Generation

GRID Model

During inference, the trained CNN-BiLSTM-CTC model checkpoint is loaded to process incoming sequences of lip frames (Figure 3.9). Each frame is passed through the CNN backbone to extract spatial features, which are then sequentially fed into the Bi-LSTM layers. The network produces a probability distribution over the character set at each time step [7].

These probabilities are decoded using a CTC beam search strategy. Beam search improves prediction accuracy by considering multiple candidate sequences and selecting the most likely transcription while removing blank tokens and merging repeated characters [8][12]. The output of the CTC decoding step is the predicted sentence corresponding to the observed lip movements.

The predicted text is forwarded to the user interface, where it is displayed in real time. The text can also be passed to a Text-to-Speech module to generate audible output, providing an assistive communication mechanism for users with speech impairments. This real-time inference pipeline ensures low-latency predictions while maintaining high accuracy for both predefined commands and general sentence-level lipreading.

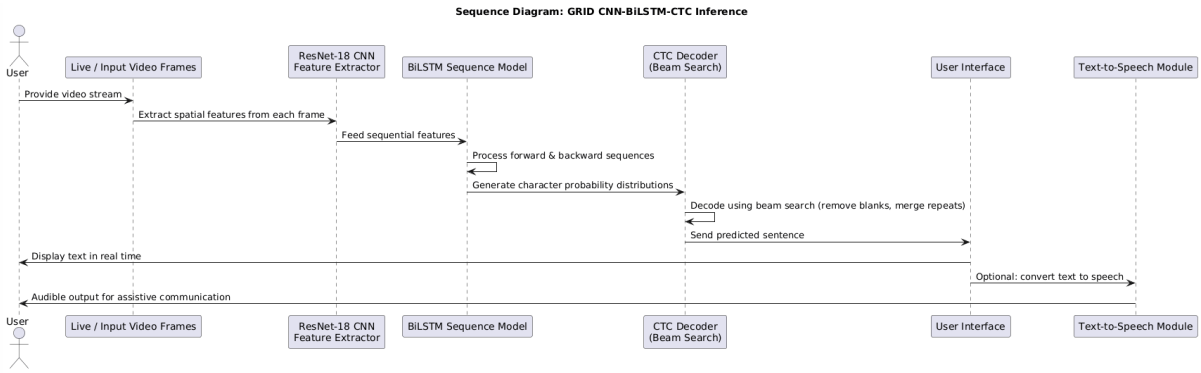


Figure 3.9: GRID Module 3

Custom Model

In the inference phase, the trained model is deployed for real-time lip reading using webcam input or pre-recorded videos. The incoming video stream is processed frame-by-frame, where MediaPipe FaceMesh detects facial landmarks and extracts the lip region of interest. The lip frames are resized, normalized, and accumulated into fixed-length sequences.

Once a sufficient number of frames is collected (Figure 3.10), the sequence is passed through the CNN–BiLSTM model. The CNN extracts spatial features from each frame, and the BiLSTM captures temporal patterns across the sequence. The final feature representation is passed through a softmax classifier, which produces probability scores for each word in the predefined vocabulary. The word with the highest probability is selected as the predicted output.

The predicted text is displayed through the user interface in real time. Additionally, a text-to-speech module can be integrated to convert the predicted word into audio output, enhancing usability for assistive communication scenarios. The system is designed to operate with low latency, enabling near real-time interaction.

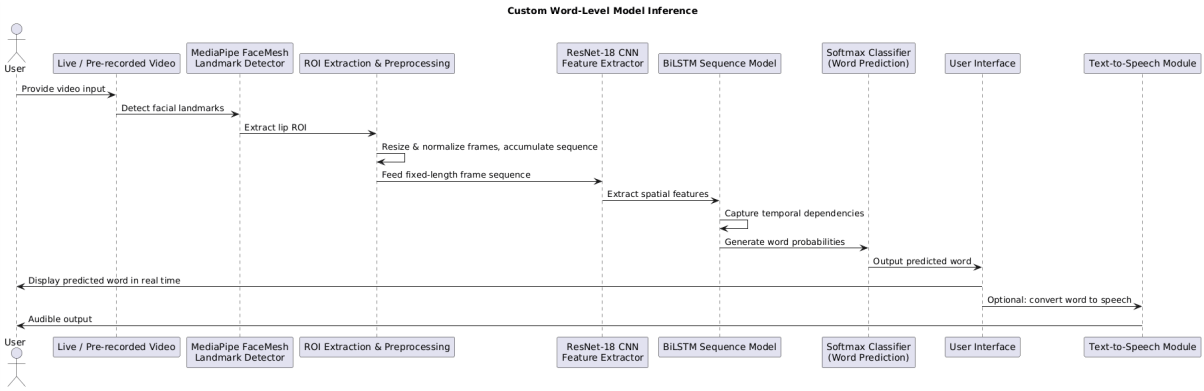


Figure 3.10: Custom Module 3

3.4 System Requirements

3.4.1 Hardware Requirements

The system is designed to run on commonly available consumer devices with moderate computational capability. On the user side, a high-resolution camera is required, which can be either a standard built-in webcam or an external camera. From the development perspective, the system requires a minimum of 8 GB RAM and an NVIDIA GeForce RTX 1650 GPU or an equivalent graphics processing unit for efficient computation and acceleration. The system is compatible with multiple device types, including smartphones, tablets, and laptops or desktop computers.

3.4.2 Software Requirements

The system is developed using Python programming language, specifically version 3.10.18. For machine learning tasks, the PyTorch framework (version 2.5.1+cu121) is utilized. Facial landmark detection is implemented using MediaPipe (version 0.10.21). For speech synthesis, the system uses pyttsx3, which is an offline text-to-speech library. Additionally, Anaconda (conda version 25.7.0) is used as the environment manager to handle dependencies and package management efficiently.

3.5 Work breakdown & Responsibilities

The system is divided into multiple functional modules, each responsible for a specific stage in the pipeline.

The **Input Capturing** module is responsible for acquiring real-time video data from the user. This module focuses specifically on capturing the mouth region using a high-resolution camera. To ensure accurate region-of-interest (ROI) detection, MediaPipe is utilized for facial landmark detection and localization. This enables the system to isolate the lip region efficiently, which is critical for downstream processing. This module is implemented by **Giribala**, who is responsible for frontend development along with providing support during the training phase.

The **Preprocessing** module handles the preparation and normalization of the captured video frames to ensure consistent input for the model. This includes extracting frames from video input, detecting facial landmarks using MediaPipe FaceMesh, and cropping the lip region of interest (ROI). The extracted lip frames are resized to a fixed resolution and converted from BGR to RGB format, as OpenCV reads images in BGR format while deep learning models and libraries expect RGB input for correct color representation. Additionally, normalization is applied to standardize pixel values across all frames. This module is developed by **Darsan**, who is responsible for backend implementation and model training.

The **Visual Speech Recognition** module is the core component of the system, where deep learning techniques are employed to interpret silent speech. This module involves designing and training models capable of analyzing lip movements and predicting corresponding phonemes, words, or complete sentences. The focus is on achieving high accuracy and real-time inference performance. This module is implemented by **Geevar**, who handles backend development and model training.

The **Output** module is responsible for presenting the recognized speech to the user in textual form. It implements a real-time transcription system that continuously updates as new predictions are generated by the model. The module ensures smooth and responsive display of output, making the system user-friendly and effective for real-time communication. This module is developed by **Fathima Meharin**, who is responsible for frontend implementation along with training support.

3.6 Key Deliverables - Output

The primary output of the system is the recognized text generated from the user's lip movements. Once the visual speech recognition pipeline processes the input video frames, the model predicts the corresponding word from a predefined vocabulary, which is then displayed as human-readable text. The system is designed to ensure that the predicted output closely reflects the user's intended speech within the supported word set.

The output is delivered in real-time or near real-time, depending on the computational resources available. During video capture, the predicted word is displayed on the user interface and updated dynamically as new lip movement sequences are processed. This enables immediate feedback, making the system suitable for applications such as silent communication and assistive technologies for individuals with speech impairments.

To achieve a seamless real-time experience, the system integrates efficient video capture, preprocessing, and inference pipelines, ensuring minimal latency between the user's lip movements and the displayed output. This responsiveness enhances usability and makes the system practical for real-world scenarios.

3.7 Project Timeline

The project is structured over a period spanning from July to February, with each phase contributing to the gradual development and completion of the system. The initial phase in July focuses on planning, where the overall project objectives, system architecture, and development approach are defined.

During August, a comprehensive literature review is conducted to understand existing research, methodologies, and technologies related to visual speech recognition and lip-reading systems. This phase helps in identifying suitable techniques and tools for implementation.

In September, the focus shifts to exploring and evaluating pre-trained models that can be leveraged or fine-tuned for the system. This is followed by data collection, where relevant datasets are gathered or prepared to support model training and evaluation.

By October, the development of the front-end setup begins, enabling user interaction and interface design. Around this stage, the project reaches approximately 30% completion, marking the transition from research to implementation.

In November, further development and integration efforts are carried out, leading to 50% completion of the project. This phase includes refining models, improving preprocessing techniques, and enhancing system components.

The month of December is dedicated to documentation efforts, including the preparation of research papers and potential patent submissions. By this time, the project reaches around 75% completion, with most core functionalities in place.

In January, the system is extended to support real-time setup, ensuring that all components work seamlessly for live input processing and output generation. Finally, in February, the project is finalized with full system integration, testing, and validation to ensure reliability and performance.

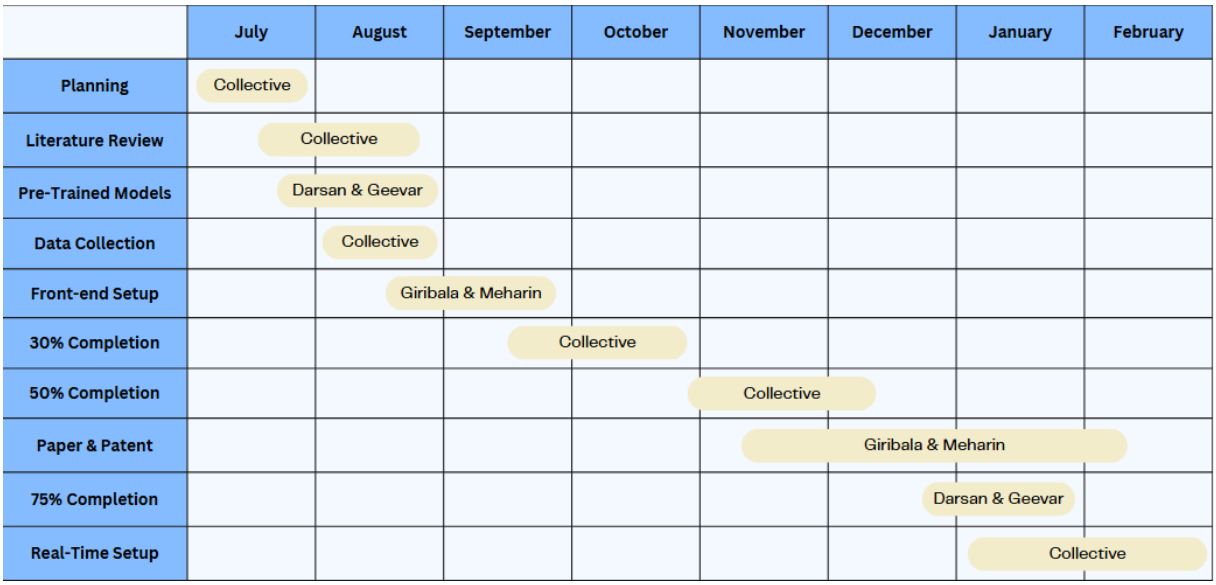
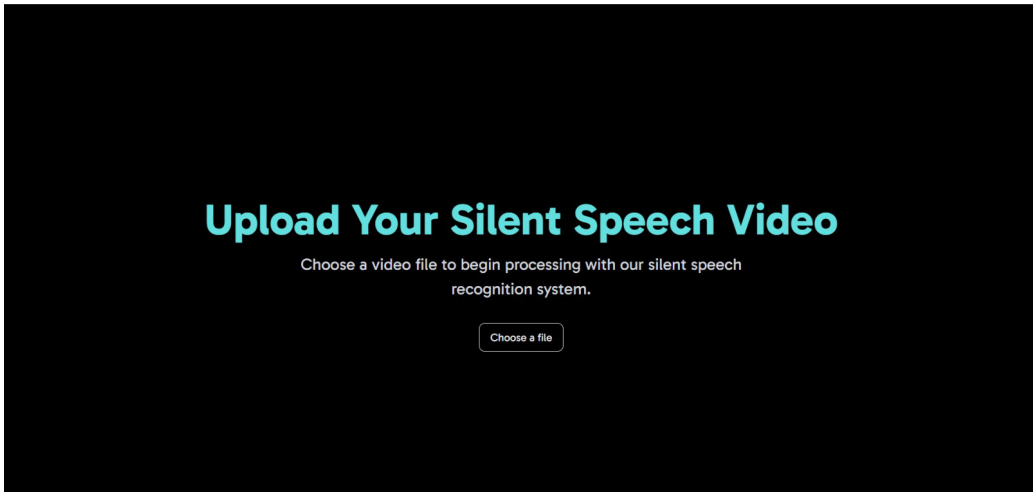
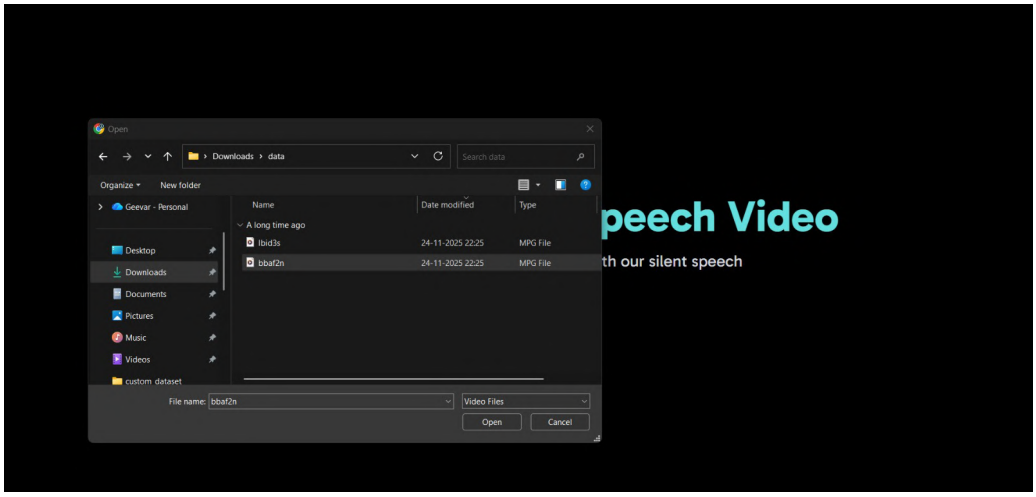
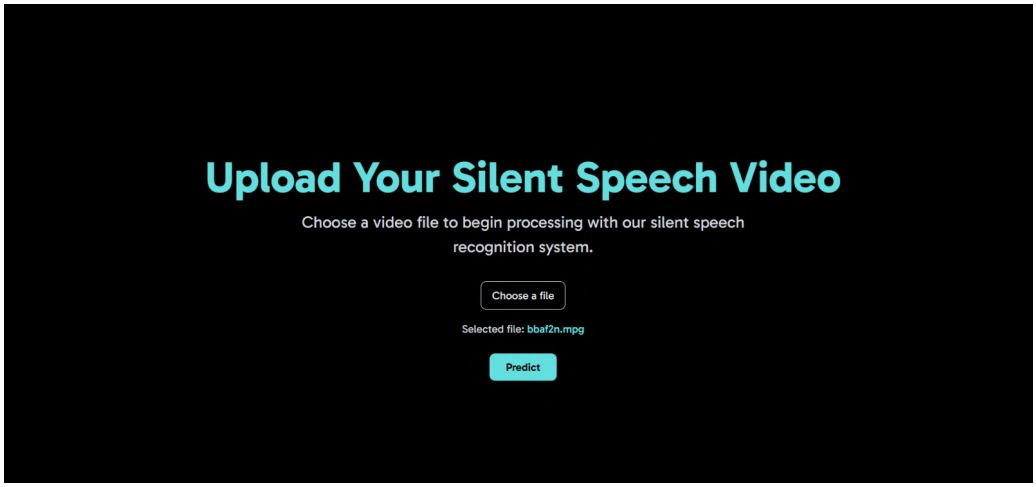


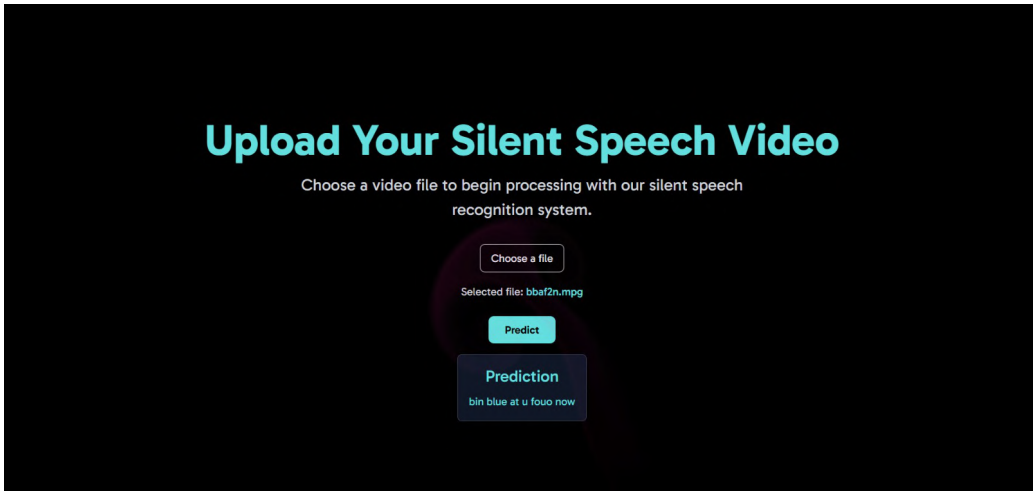
Figure 3.11: Gantt Chart

3.8 User Interface Implementation








Upload Your Silent Speech Video
Choose a video file to begin processing with our silent speech recognition system.
Choose a file
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Predict
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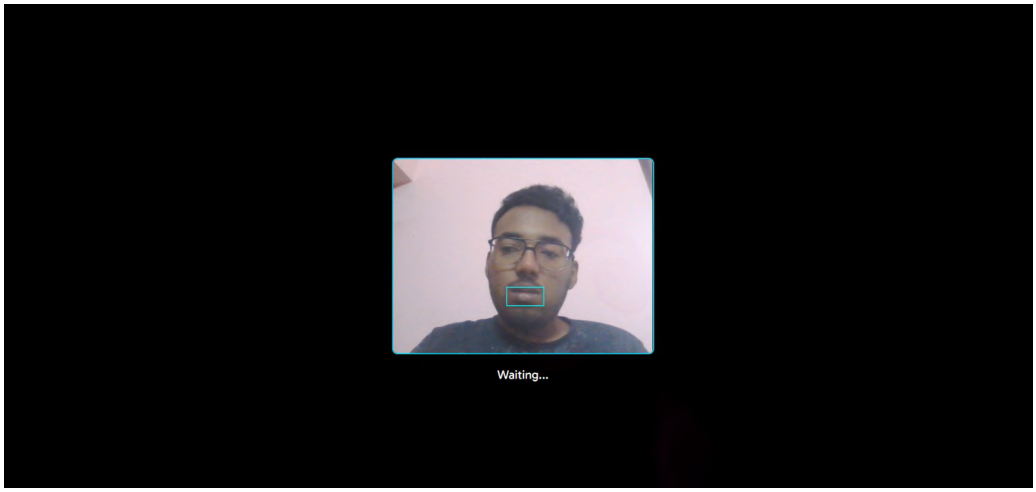
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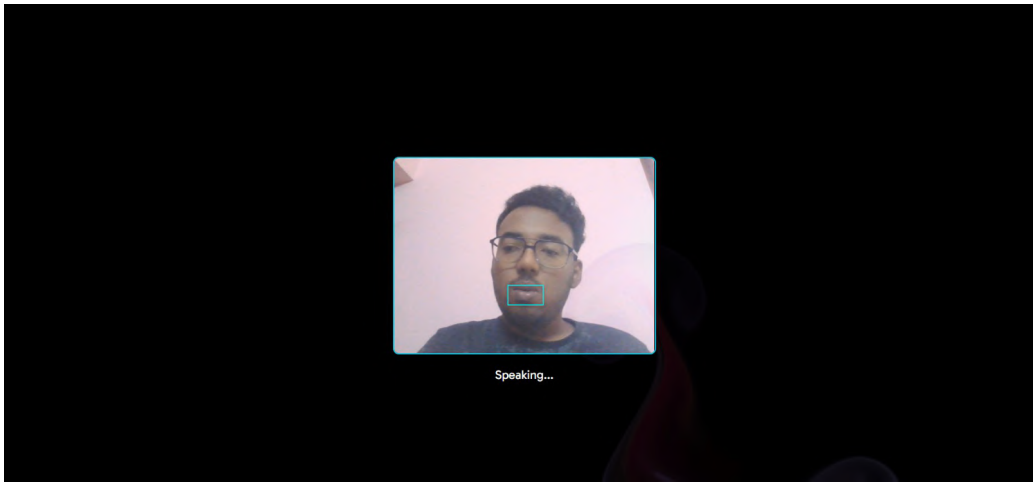
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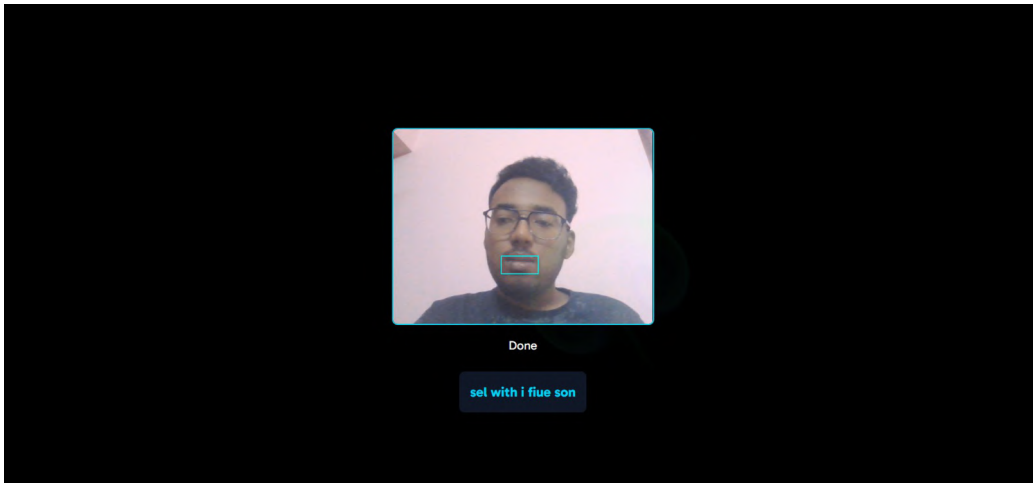

3.9 Risks & Challenges

The system faces several challenges that can impact its overall performance and reliability. One of the primary issues is related to lighting conditions. Poor or inconsistent lighting can significantly reduce the accuracy of mouth detection and recognition, as the system relies heavily on clear visual input to identify lip movements. Variations in illumination may lead to loss of important features.

Another major challenge is the variability in lip movements among different users. Individuals have unique speaking styles, articulation patterns, and lip shapes, which makes it difficult for a generalized model to perform consistently. This variability can reduce prediction accuracy, especially when the model encounters unseen speaking patterns.

Head movements during video capture also pose a significant problem. Excessive or rapid movements can disrupt the tracking of facial landmarks, leading to incorrect ROI extraction. This affects the quality of input provided to the recognition model and may result in inaccurate predictions.

The availability of datasets is another limitation. There is a lack of large-scale datasets, which restricts the ability to train robust and highly accurate deep learning models.

Finally, the system requires substantial computational resources due to the use of deep learning models. Real-time processing demands powerful hardware, such as GPUs, to ensure low latency and efficient performance. This can make deployment challenging on devices with limited computational capacity.

3.10 Summary of Chapter

This chapter presented the complete system design for the proposed visual speech recognition framework. The overall architecture integrates input acquisition, preprocessing, model training, and output generation to enable accurate real-time lipreading. The data acquisition module captures video streams or archival files and extracts the lip region of interest (ROI) using MediaPipe FaceMesh, followed by preprocessing steps such as normalization, resizing, and temporal sequencing.

Feature extraction and model training utilize a hybrid CNN–BiLSTM–CTC architecture. The ResNet-18 CNN backbone extracts spatial features from lip frames, while the BiLSTM captures temporal dependencies across sequences. Connectionist Temporal Classification (CTC) enables alignment-free decoding for sentence-level predictions, and softmax classification is applied for custom word-level datasets. The training process employs backpropagation with the AdamW optimizer to ensure efficient convergence and generalization.

During inference, the system produces real-time textual output from lip movements, which can also be converted to speech using a text-to-speech engine. The work breakdown defines module responsibilities across team members, ensuring a coordinated development workflow. A project timeline outlines sequential milestones, from planning and literature review to dataset preparation, model development, front-end integration, and final real-time deployment.

The chapter also highlighted key challenges, including lighting variations, user-specific articulation differences, head movements, dataset limitations, and computational requirements. Overall, the design emphasizes modularity, real-time performance, and accuracy, providing a robust foundation for implementing a practical visual speech recognition system.

Chapter 4

Implementation

This chapter describes the practical implementation of the proposed silent speech recognition system. The system was developed using Python and deep learning frameworks to convert lip movements into text. The implementation follows the architecture described in Chapter 3 and includes dataset preparation, video preprocessing, model training, and real-time inference.

4.1 Algorithms Used

The proposed visual speech recognition system employs several advanced algorithms, each playing a critical role in the pipeline. These algorithms are responsible for facial landmark detection, spatial and temporal feature extraction, sequence modeling, classification, and optimization.

4.1.1 MediaPipe FaceMesh Algorithm

MediaPipe FaceMesh is utilized for accurate facial landmark detection and extraction of the lip region of interest (ROI). It leverages a deep learning–based approach to detect 468 facial landmarks. Specific landmarks around the lips are used to define a bounding box, enabling precise cropping of the mouth region for further processing.

Procedure:

1. Capture an input video frame.
2. Detect 468 facial landmarks using MediaPipe FaceMesh.
3. Select landmark points corresponding to the lips.
4. Compute a bounding box around the lip landmarks.
5. Crop the lip region to extract the ROI for downstream processing.

4.1.2 Convolutional Neural Network (CNN – ResNet-18)

A Convolutional Neural Network (CNN) based on the ResNet-18 architecture is used for spatial feature extraction from cropped lip frames. The network applies convolutional filters to capture local patterns, uses ReLU activation functions to introduce non-linearity, and employs pooling operations to reduce dimensionality while retaining essential visual features such as lip shape and texture.

Procedure:

1. Input cropped lip frame (size 64×64 pixels).
2. Apply convolutional filters to extract spatial features.
3. Apply ReLU activation to introduce non-linearity.
4. Perform pooling operations to reduce dimensionality.
5. Flatten or aggregate features to form a feature vector for each frame.

4.1.3 Bidirectional Long Short-Term Memory (BiLSTM)

BiLSTM networks are used to capture temporal dependencies across sequences of lip frames. Unlike standard LSTMs, BiLSTMs process sequences in both forward and backward directions, allowing the model to incorporate context from past and future frames. Gating mechanisms (input, forget, and output gates) regulate information flow, enabling retention of important temporal features while discarding irrelevant information.

Procedure:

1. Input a sequence of feature vectors extracted by the CNN.
2. Process the sequence in both forward and backward directions.
3. Apply LSTM gating mechanisms to control memory and information flow.
4. Capture temporal dependencies and aggregate contextual features.
5. Output a sequence representation for subsequent decoding.

4.1.4 Connectionist Temporal Classification (CTC)

CTC is employed for sequence prediction on the GRID dataset. It allows the system to map input sequences to output text without requiring explicit alignment between video frames and textual labels. CTC efficiently handles variable-length sequences and repetitive or blank frames.

Procedure:

1. Receive sequential outputs from the BiLSTM network.
2. Compute probability distributions over the character set for each timestep.
3. Remove repeated predictions to consolidate duplicates.
4. Remove blank tokens representing no prediction.
5. Generate the final textual output.

4.1.5 Softmax Classification Algorithm

For the custom dataset, a softmax-based classifier is used to predict predefined word classes. The softmax layer converts raw network outputs into probabilities, allowing the system to select the most probable class for each input sequence.

Procedure:

1. Input the feature representation from the BiLSTM or CNN.
2. Apply a fully connected layer to map features to class logits.
3. Compute softmax probabilities across all word classes.
4. Select the class with the highest probability.
5. Output the predicted word.

4.1.6 Optimization Using AdamW

The model is trained using backpropagation with the AdamW optimizer, which combines adaptive learning rates with weight decay regularization. This allows efficient convergence

while preventing overfitting, resulting in improved generalization performance.

Procedure:

1. Compute the loss between predicted and target outputs.
2. Perform backpropagation to compute gradients.
3. Update network weights using the AdamW optimizer.
4. Apply weight decay to prevent overfitting.
5. Repeat the process over multiple epochs until convergence.

4.2 Datasets Used

Two datasets were used in this project: the GRID dataset and a custom lip-reading dataset.

4.2.1 GRID Dataset

The GRID dataset was utilized to train the sentence-level visual speech recognition model. It contains audiovisual recordings of multiple speakers, with each speaker uttering sentences following a fixed grammatical structure consisting of six components: command, color, preposition, letter, digit, and adverb [8]. For this project, only the visual stream of the videos was utilized, while the audio component was ignored.

Each video sequence was processed to extract individual frames. Facial landmarks were detected using MediaPipe FaceMesh, and the lip region was localized using landmark points around the mouth. The extracted lip region from each frame was cropped and resized to a fixed spatial resolution suitable for model training [9][10]. The processed frames were stored in structured directories corresponding to each video sample.

To efficiently manage the dataset during training, manifest files were generated. Each entry contains the sample identifier, the directory containing the extracted lip frames, and the corresponding transcript obtained from the alignment files provided with the GRID dataset. Multiple speaker-specific manifest files were merged into a single consolidated manifest used by the training pipeline [1].

4.2.2 Custom Emergency Word Dataset

To improve real-time performance and support emergency communication scenarios, a custom dataset was collected. This dataset focuses on recognizing a small vocabulary of emergency-related words: *Yes, No, Help, Pain,* and *Water*. Recordings were obtained from multiple speakers using webcams and smartphone cameras under varying environmental conditions, including different lighting levels and speaking speeds.

The videos were processed using the same preprocessing pipeline as the GRID dataset. Lip frames were extracted, resized, and stored as ordered frame sequences. A separate manifest file was created for the custom dataset, containing the directory of extracted frames and the corresponding word labels.

The dataset was divided into training and testing subsets to evaluate the fine-tuned word-level classifier and validate system performance in real-time operating conditions.

4.2.3 Dataset Overview

Table 4.1 summarizes the key characteristics of both datasets used in this project.

Table 4.1: Summary of Datasets Used

Feature	**GRID Dataset**	**Custom Emergency Dataset**
Purpose	Sentence-level VSR	Word-level emergency recognition
Vocabulary Size	Fixed grammar sentences	5 words (Yes, No, Help, Pain, Water)
Number of Speakers	34	10+
Video Source	Professional recordings	Webcam / Smartphone
Frame Rate	25 fps	30 fps
Preprocessing	Lip extraction + resizing	Lip extraction + resizing
Training/Test Split	90:10	80:20

4.3 Video Preprocessing

Video preprocessing is performed to prepare input data for the deep learning model. The main goal is to extract and normalize the lip region while preserving the temporal dynamics of speech.

4.3.1 Frame Extraction and Color Conversion

Video frames are first extracted from each video using the OpenCV library. These frames represent the temporal sequence of lip movements for each utterance. Since OpenCV reads images in BGR format and MediaPipe FaceMesh expects RGB input, a BGR-to-RGB conversion is performed before facial landmark detection [1].

4.3.2 Facial Landmark Detection and ROI Extraction

MediaPipe FaceMesh is used to detect 468 facial landmark points in each frame. A subset of these points corresponding to the lips is used to compute a bounding box around the mouth region. This isolates the Region of Interest (ROI) containing the lip movements, allowing the model to focus only on relevant visual features.

4.3.3 Cropping, Resizing, and Normalization

The extracted lip region is cropped and resized to a fixed spatial resolution of 64×64 pixels to match the input size required by the CNN–BiLSTM network. Each lip frame is then converted into a tensor representation and normalized to have pixel values in a consistent range, ensuring stable model training [9].

4.3.4 Sequence Preparation

Processed frames are arranged into fixed-length temporal sequences [10]. To maintain consistent sequence lengths across all samples, shorter sequences are padded while longer sequences are truncated. This ensures that the temporal structure of lip movements is preserved and that all sequences can be efficiently processed in batches by the model.

4.3.5 Preprocessing Parameters

Table 4.2 summarizes the key preprocessing parameters used for video and lip frame preparation.

4.4 Sentence-Level Model Implementation

The sentence-level lip-reading model was implemented using the PyTorch deep learning framework. It combines a convolutional neural network for spatial feature extraction with

Table 4.2: Video Preprocessing Parameters

Parameter	**Value**
Frame Extraction	OpenCV
Color Conversion	BGR to RGB
Facial Landmark Detection	MediaPipe FaceMesh (468 points)
Lip ROI Resolution	64×64 pixels
Sequence Length	Fixed; padding/truncation applied
Normalization	Pixel values scaled to [0,1]

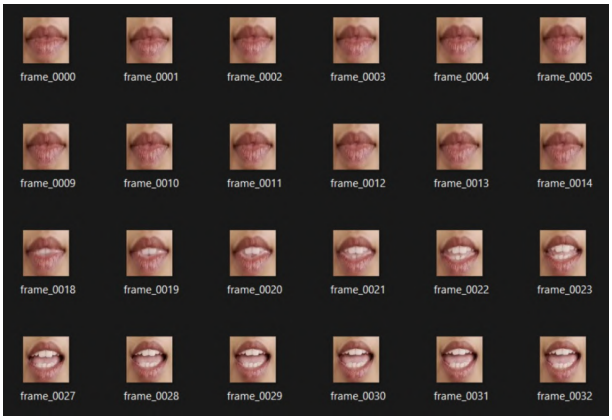

Figure 4.1: Example sequence of cropped lip frames extracted from a video using MediaPipe FaceMesh.

a recurrent network for temporal modeling, enabling end-to-end silent speech recognition.

4.4.1 Frame-Level Feature Extraction

A pre-trained ResNet-18 network was employed as the frame-level feature extractor. The final classification layer of ResNet-18 was removed, and the convolutional layers were frozen to preserve the visual features learned during ImageNet pretraining while reducing the number of trainable parameters [1][7]. Each input lip frame is processed by the ResNet backbone and projected into a 256-dimensional feature vector, capturing essential visual patterns of the lips.

4.4.2 Temporal Modeling with BiLSTM

The frame-level feature vectors are arranged as a temporal sequence and passed to a two-layer Bidirectional Long Short-Term Memory (BiLSTM) network with a hidden size of 256. The BiLSTM captures temporal dependencies between consecutive frames, modeling the dynamics of lip movements over time. This bidirectional architecture allows the network to incorporate both past and future context when making predictions.

4.4.3 Sequence Alignment and Decoding

To handle the alignment between variable-length frame sequences and text labels, Connectionist Temporal Classification (CTC) loss was used during training. CTC allows the model to learn the mapping between frame sequences and character-level transcripts without requiring exact frame-level alignment. During inference, greedy CTC decoding was applied to generate the predicted sequences from the model outputs [12].

4.4.4 Model Architecture

Figure 4.2 illustrates the overall architecture of the CNN–BiLSTM–CTC sentence-level lipreading model. Each input frame is first processed by the ResNet-18 feature extractor, followed by temporal modeling with a BiLSTM network. The final sequence is decoded using CTC to produce the transcription [1][7].

4.4.5 Training Configuration

The model was trained on the GRID dataset using the AdamW optimizer with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}. Training was performed for 50 epochs with a batch size of 4. During validation, predictions were generated using greedy CTC decoding, and performance was measured using Character Error Rate (CER). The checkpoint achieving the lowest validation CER was saved as the final model.

4.5 Word-Level Model Fine-Tuning

To support real-time recognition of predefined commands, the trained sentence-level model was fine-tuned using a custom word-level dataset. Transfer learning was applied by initializing the model with weights learned from the GRID sentence-level model, allowing

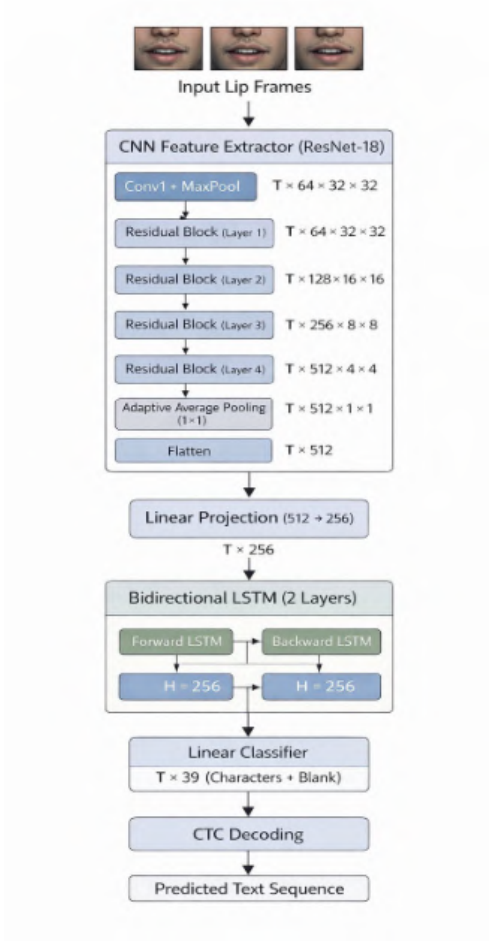

Figure 4.2: Architecture of the proposed CNN-BiLSTM-CTC silent speech recognition model.

reuse of previously learned spatial and temporal lip movement features.

The custom dataset consists of short video clips corresponding to individual words or commands, enabling the model to adapt from sentence-level sequence prediction to word-level classification. During fine-tuning, the convolutional backbone (ResNet-18) retains its ability to extract spatial lip features, while the temporal modeling layers (BiLSTM) are further optimized to capture shorter temporal dependencies associated with isolated word utterances.

To ensure efficient adaptation, most of the convolutional layers are kept frozen, while selected higher layers and the classification head are fine-tuned. This approach preserves general learned features while adapting the model to the custom vocabulary, reducing training time and improving generalization on a smaller dataset.

Unlike the sentence-level model, which uses CTC decoding, the word-level model employs

a softmax-based classifier to directly predict predefined word classes. This simplifies the prediction task and improves performance for fixed vocabulary recognition.
The fine-tuned model is evaluated using classification accuracy on the custom test dataset, demonstrating effective recognition of command-based inputs.This fine-tuning process enables real-time operation, allowing word-level prediction for assistive communication applications.

4.5.1 Architecture Modifications

For the word-level task, the original CTC output layer was replaced with a fully connected classification layer. The CNN backbone (ResNet-18) and BiLSTM layers remain unchanged from the sentence-level model. To aggregate temporal features into a single representation, a temporal mean pooling operation was applied [1][7].
During fine-tuning, most layers of the ResNet backbone were frozen to preserve visual features learned from the larger GRID dataset [5].

4.5.2 Training Configuration

The model was trained using cross-entropy loss with the AdamW optimizer and a learning rate of 1×10^{-4}. The training hyperparameters used for the custom model and GRID model are summarized in the Table 4.3.

Table 4.3: Training Hyperparameters

Hyperparameter	**Custom Model**	**GRID Model**
Optimizer	AdamW	AdamW
Learning Rate and Weight Decay	1×10^{-4}	1×10^{-4}
Batch Size	8	4
Epochs	25	75
Frame Feature Dimension	256	256

4.6 Real-Time Inference System

A real-time inference pipeline was developed to enable silent speech recognition from live webcam input or uploaded video clips. In this pipeline, video frames are first captured

from a webcam using OpenCV. MediaPipe FaceMesh is then applied to detect facial landmarks in each frame, after which the lip region is cropped and resized to 64×64 pixels. The processed frames are stored in a fixed-length buffer of up to 75 frames. The model generates class probabilities via a softmax layer, and a confidence threshold is applied to filter uncertain predictions [13]. The predicted word is displayed on the user interface, and optionally, the output text can be converted into speech using a Text-to-Speech (TTS) [3] module to assist users with speech impairments.
Table 4.4 summarizes the key configurations and performance metrics for both fine-tuning and real-time inference of the word-level model.

Table 4.4: Detailed Summary of Word-Level Model Fine-Tuning and Real-Time Pipeline

Component	**Configuration / Description**
Model Architecture	Based on CNN-BiLSTM-CTC sentence-level model. Fine-tuned by replacing the CTC layer with a fully connected classification layer for a vocabulary of **5 emergency words: Yes, No, Help, Pain, Water**. CNN and BiLSTM layers retained. Temporal mean pooling applied across frames.
Training Setup	**Loss:** Cross-Entropy. **Optimizer:** AdamW, **Learning Rate:** 1×10^{-4}. **Batch Size:** 8. **Max Sequence Length:** 75 frames. Evaluated on held-out test set; checkpoint with highest validation accuracy used for deployment.
Real-Time Inference Pipeline	Frames captured from webcam (OpenCV) or uploaded videos. **Landmark Detection:** MediaPipe FaceMesh (468 points). Lip region cropped and resized to 64×64 pixels. Buffered sequence passed to model; softmax probabilities computed. Confidence threshold applied for reliable predictions. Optional Text-to-Speech (TTS) output.
Performance Metrics	**Test Accuracy:** 87.61%. **Average Latency:** 0.4–0.5 seconds per prediction. Robust performance under varying lighting and speaking speeds.

Chapter 5

Results and Discussions

This chapter presents the experimental results and performance evaluation of the proposed Silent Speech Recognition system. The system was evaluated using two datasets: the GRID dataset for sentence-level lipreading and a custom emergency vocabulary dataset for word-level recognition. The performance of the model was analyzed using evaluation metrics such as Character Error Rate (CER), character accuracy, and classification accuracy. In addition, the effectiveness of the real-time system using webcam input was also examined.

5.1 Evaluation of Sentence-Level Model

The sentence-level lipreading model was trained and evaluated using the GRID audiovisual speech dataset, which is a widely used benchmark dataset for audio-visual speech recognition and lipreading research. The dataset contains recordings of multiple speakers uttering fixed grammar-based sentences, allowing models to learn the relationship between lip movements and spoken language. Each sentence in the dataset follows a predefined structure, making it suitable for evaluating sentence-level visual speech recognition systems.

For training and evaluation, the dataset was divided into training and validation sets using a 90:10 split. The training set was used to learn the model parameters, while the validation set was used to evaluate the generalization capability of the model and monitor performance during training.

The proposed model architecture combines convolutional and recurrent neural networks to capture both spatial and temporal information from video sequences. A Convolutional Neural Network (CNN) is first used to extract spatial features from the cropped mouth region in each frame. These features represent important visual patterns such as

lip shape, movement, and articulation cues. The extracted frame-level features are then passed to a Bidirectional Long Short-Term Memory (Bi-LSTM) network, which models the temporal dependencies across consecutive frames. Since lip movements evolve over time during speech production, the Bi-LSTM enables the model to capture both past and future contextual information in the sequence.

The output of the Bi-LSTM network is fed into a Connectionist Temporal Classification (CTC) layer for sequence prediction. The CTC layer allows the model to generate sentence-level predictions without requiring explicit alignment between input video frames and output text sequences. This is particularly useful for lipreading tasks where the timing of individual characters or phonemes may vary across speakers.

The performance of the model was evaluated using the Character Error Rate (CER), which is a commonly used metric in speech recognition and lipreading systems. CER measures the difference between the predicted text sequence and the ground-truth transcript at the character level.

It is defined as:

$$CER = \frac{S + D + I}{N} \tag{5.1}$$

where S represents the number of substitutions, D represents the number of deletions, I represents the number of insertions, and N denotes the total number of characters in the reference transcript. A lower CER value indicates better recognition performance and higher accuracy of the predicted sentence.

The evaluation results obtained from the GRID dataset are presented in Table 5.1. The results demonstrate that the CNN-BiLSTM-CTC architecture is capable of effectively learning the spatial and temporal patterns associated with lip movements. The CNN component extracts meaningful visual features from each frame, while the Bi-LSTM captures the temporal relationships between these features across the entire sequence.

Furthermore, the relatively low CER obtained during evaluation indicates that the model can successfully generate accurate sentence-level predictions from silent lip movements. This highlights the effectiveness of combining convolutional feature extraction with recurrent sequence modeling for visual speech recognition tasks. However, variations in speaker articulation, lighting conditions, and subtle lip movements may still introduce recognition

errors, suggesting opportunities for further improvements through attention mechanisms, larger datasets, or multimodal integration.

Table 5.1: Performance of Sentence-Level Classifier

Metric	**Result**
Vocabulary Size	39 Characters
Maximum Sequence Length	75 Frames
Batch Size	4
Test Accuracy	93.93%

5.2 Evaluation of Word-Level Classifier

To support real-time emergency communication scenarios, the trained sentence-level lipreading model was fine-tuned to perform word-level classification on a custom dataset. This task focuses on recognizing a small set of critical emergency-related words, namely: *Yes, No, Help, Pain, and Water.* Such a constrained vocabulary is particularly useful in assistive communication systems where fast and reliable recognition of key terms is more important than full sentence generation.

Unlike the sentence-level model, which generates character sequences using a CTC-based decoding mechanism, the fine-tuned model performs direct classification. The final sequence representation obtained from the Bi-LSTM layer is passed through a fully connected layer followed by a softmax activation function. The class with the highest probability is selected as the predicted word.

During fine-tuning, the lower layers of the model (CNN and Bi-LSTM) retain the learned feature representations from the GRID dataset, while the final classification layer is adapted to the new vocabulary. This transfer learning approach allows the model to leverage previously learned spatio-temporal features of lip movements, reducing the amount of data required for training the new task.

The dataset was divided into training and testing subsets, and the model was evaluated on the held-out test set to assess its generalization performance. Table 5.2 presents the evaluation results of the word-level classifier.

The results indicate that transfer learning from the sentence-level GRID model signifi-

cantly improves classification performance, even when trained on a relatively small custom dataset. The achieved test accuracy of 87.61% demonstrates that the model can effectively learn discriminative lip movement patterns associated with the predefined emergency words.

Furthermore, the high accuracy suggests that the extracted visual features generalize well across tasks, from sentence-level prediction to word-level classification. However, some misclassifications may still occur due to similarities in lip movements between certain words, as well as variations in speaking style, speed, and lighting conditions. Future improvements could include expanding the dataset, incorporating attention mechanisms, or integrating audio signals to enhance robustness in real-world scenarios.

Table 5.2: Performance of Word-Level Classifier

Metric	**Result**
Vocabulary Size	5 Words
Maximum Sequence Length	75 Frames
Batch Size	8
Test Accuracy	87.61%

5.3 Real-Time System Performance

The trained word-level classifier was integrated into a real-time system using webcam input. The system captures video frames and detects facial landmarks using MediaPipe FaceMesh. The lip region is extracted from each frame and processed as a sequence for prediction.

When sufficient frames are collected, the frame sequence is passed to the CNN-Bi-LSTM classifier to predict the corresponding word. A confidence threshold is applied to filter unreliable predictions. If the prediction confidence is below the threshold value, the output is labeled as *Unknown*.

Experimental observations indicate that the system performs efficiently under normal webcam conditions. The model can recognize predefined emergency commands with low latency and stable performance. Table 5.3 summarizes the real-time system evaluation metrics.

The results demonstrate that the proposed system is capable of performing silent speech recognition using lip movements and deep learning techniques. The integration of the trained model with real-time webcam input makes the system suitable for assistive communication applications, particularly for individuals with speech impairments.

In summary, the experimental results demonstrate that the proposed silent speech recognition system achieves reliable performance in both sentence-level and word-level tasks. The integration of deep learning models with real-time webcam processing enables practical deployment for assistive communication systems.

Table 5.3: Real-Time System Performance Metrics

Metric	**GRID**	**Custom**
Accuracy	70-75%	40-50%
Latency	2-3 seconds	1-2 seconds
Frames per second	15-20	15-20

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

This project presented a dual-mode silent speech recognition system capable of converting visual lip movements into readable text and synthesized speech when required. The proposed system integrates MediaPipe-based facial landmark detection, a Convolutional Neural Network (CNN) for spatial feature extraction, and a Bidirectional Long Short-Term Memory (Bi-LSTM) network for temporal modeling of lip movements. By combining spatial and temporal learning techniques, the system can interpret silent lip movements and generate meaningful textual outputs.

In this work, two complementary models were developed. A sentence-level CNN-Bi-LSTM-CTC model trained on the GRID dataset achieved a Character Error Rate (CER) of 0.0607, corresponding to a character recognition accuracy of 93.93%. The learned visual representations were then adapted through transfer learning to a word-level classifier trained on a custom emergency vocabulary dataset consisting of five words. The fine-tuned model achieved a test accuracy of 87.61%, demonstrating the effectiveness of transfer learning even with limited training data. The trained classifier was further integrated into a real-time inference system using webcam input, enabling recognition of predefined commands and displaying the predicted text to the user. These results demonstrate the feasibility of practical silent speech recognition systems for assistive communication applications.

6.2 Future Scope

Future work will focus on extending the proposed vision-based silent speech recognition system toward a wearable sensing platform. The current camera-based approach may be affected by environmental factors such as lighting conditions and head pose varia-

tions. To overcome these limitations, future research can explore a jaw–cheek muscle sensing wearable capable of detecting silent speech using non-contact sensors. Integrating such sensor-based signals with machine learning models could enable a more portable, privacy-preserving, and robust silent communication interface. Additionally, expanding the custom dataset by increasing the vocabulary size and collecting recordings from a larger number of speakers may further improve the generalization capability and accuracy of the system.

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Appendix A: Presentation

1

100% EVALUATION PRESENTATION

SILENT SPEECH RECOGNITION FOR THE VOICELESS

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Introduction

Silent Speech Recognition enables communication through facial and mouth movements, offering a solution for individuals with speech impairments.

- **Growing Need**: Millions face speech loss, leading to isolation and distress.
- **Current Limitations**: Existing devices often require manual operation, unsuitable for users with motor impairments.
- **Technological Opportunity**: Computer vision enable hands-free, real-time speech recognition.
- **Empowerment Through Technology**: SSR offers a potential breakthrough by providing a fully hands-free, natural, and real-time method of communication.

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Problem Definition

Existing AAC systems are unsuitable for individuals with both speech and motor impairments, as they rely on manual input and fail to provide natural, real-time communication. There is a need for a non-invasive, vision-based solution that can accurately interpret silent speech.

Purpose & Need

- Enable people with speech impairments to **communicate using facial and mouth movements** through AI and computer vision.
- Provide a **natural, hands-free, and real-time communication** method.
- Restore independence and confidence for those unable to speak due to illness or disability.
- Address emotional distress and social isolation caused by speech loss.
- Overcome the limitations of existing aids that require manual operation.
- Utilize advancements in AI and computer vision for real-time, touch-free speech recognition.

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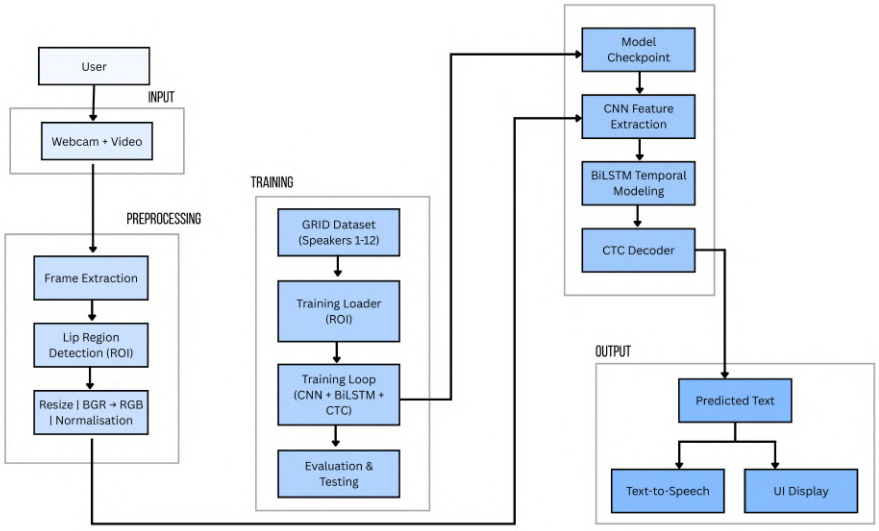
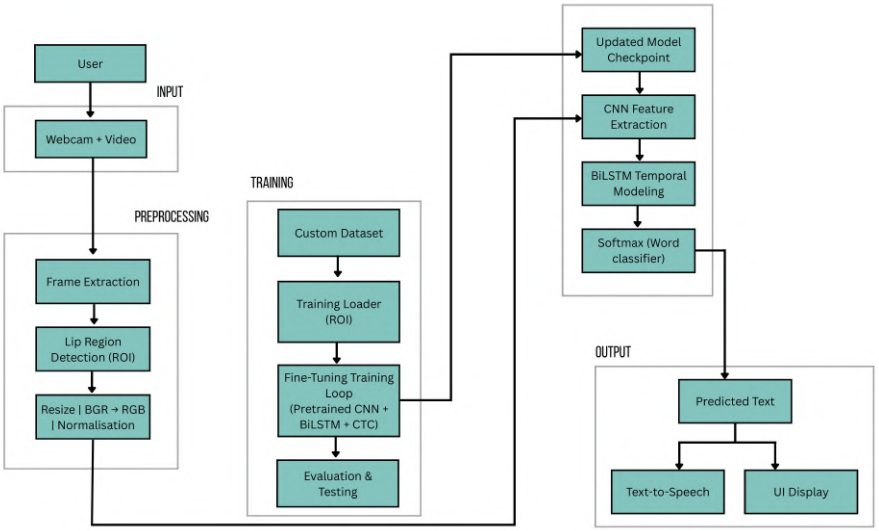
Objectives

- **To Capture Facial and Mouth Movements**: Develop a system to detect and track lip movements to interpret speech.
- **To Implement Real-Time Processing**: Use computer vision algorithms to process and decode movements in real-time, ensuring smooth communication.
- **To Create Hands-Free Interaction**: Design a fully hands-free interface for users, eliminating the need for manual input.
- **To Enhance Accessibility**: Ensure the system is intuitive, adaptable, and usable for individuals with motor impairments or other disabilities.

Paper	Main Focus	Model Type	Key Techniques	Applications
Audio-Visual Speech Recognition in Noisy Environments [10]	Combining audio and visual inputs for speech recognition	CNNs, LSTMs, AVSR hybrid	• CNNs for feature extraction • LSTMs for temporal modeling • Audio-Visual fusion	Robust to noisy environments, Utilizes both auditory and visual cues
Lip Reading Using Alternating 2D-3D Convolutions [11]	Lipreading system using deep learning and convolutional models	ALSOS + ResNet CNN	• 2D-3D convolutions • ALSOS • ResNet	Improved recognition performance through spatiotemporal feature extraction
Visual Speech Recognition Using LSTM Networks [12]	Deep learning models for lipreading with temporal modeling	LSTM-based VSR	• LSTMs • BiLSTMs • 3D CNNs	Strong handling of temporal dependencies
Lipreading with CTC-Based Visual Speech Recognition [13]	Sentence-level Visual Speech Recognition (VSR)	CTC-based VSR model	• CTC loss function • Spatial attention • Bidirectional recurrent layers	Direct video-to-text mapping without frame-level alignment





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Module Explanation

Input Module:
- Captures input from both live camera feed and video upload.
- Video upload pipeline is working well. Live video stream is accepted but currently slower than expected.

Preprocessing Module:
- Extracts frames from video input.
- Crops and resizes frames to standard dimensions required by the DL module.
- Fully tested with video upload input; works reliably with minimal errors.

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Module Explanation

Training Module:
- Training performed using the GRID Dataset and additional custom speaker recordings. Successfully trained on GRID subjects S1 to S12 and 11 custom speakers.
- Dataset preparation, labeling, and train-test split were completed for both datasets.
- The training pipeline is fully functional and produces consistent results.

Deep Learning Module:
- The deep learning model was trained and evaluated using both GRID dataset samples and custom speaker data.

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Module Explanation

Output Module:
- Displays predicted text from the Deep Learning module.
- Works reliably for video upload input.
- Real-time display for live video feed not fully optimized; latency is noticeable.

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Assumptions

- Lips are clearly visible and unobstructed in all video inputs.
- Only one speaker is present per video.
- Video quality (lighting, resolution) is consistent across the dataset.
- Dataset annotations are accurate and properly synchronized with the video frames.
- The model is trained and tested on GPU-enabled setup for optimal performance.

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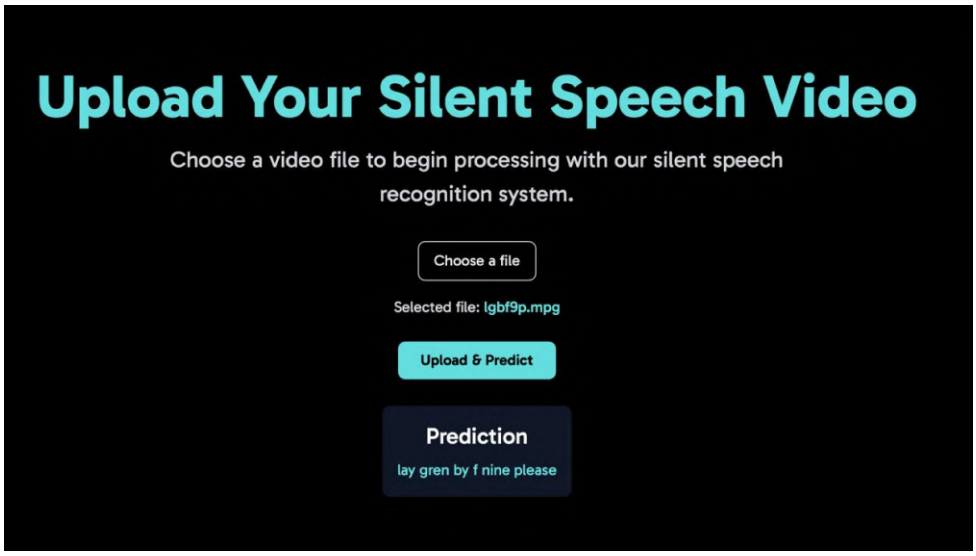
1. GRID Corpus
Details:
- Speakers: 34 (18 male, 16 female)
- Sentences per speaker: 1,000
  - <COMMAND> <COLOR> <PREPOSITION> <LETTER> <DIGIT> <ADVERB>
- Video format: .mpg (MPEG-1)

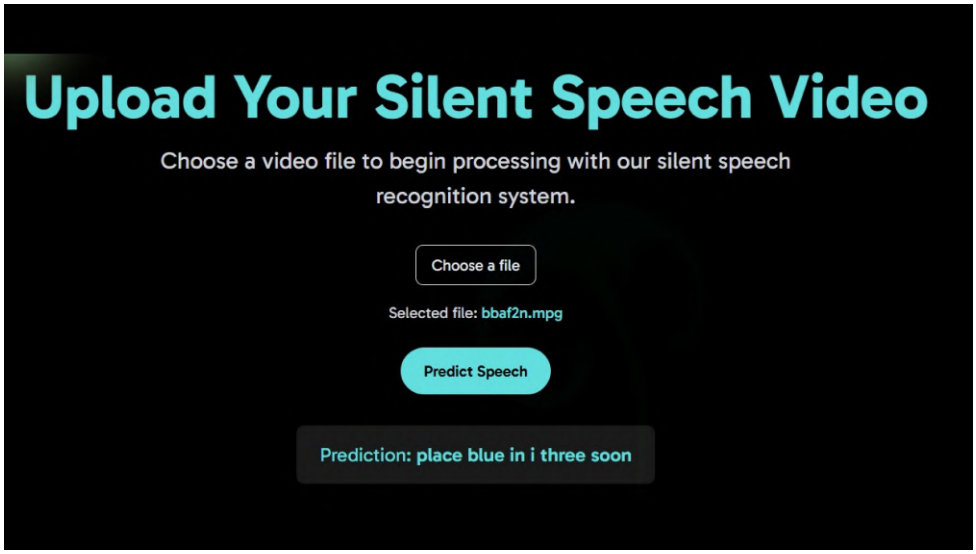
It uses a controlled vocabulary with a fixed 6-word grammatical format, which makes it easy to preprocess and align video frames with their corresponding text labels.

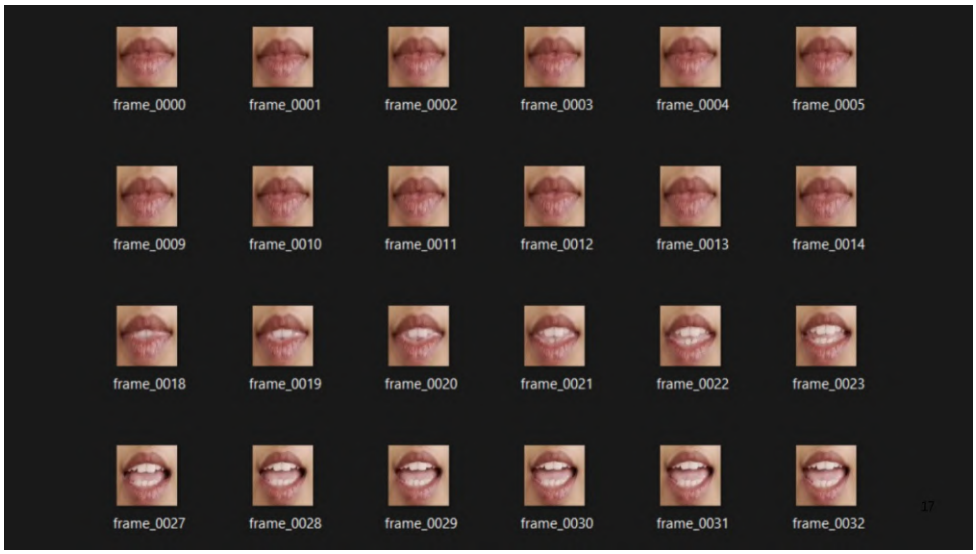
2. Custom Dataset
Details:
- Speakers: 11 (6 male, 5 female) (will be adding more speakers)
- Words per speaker: 5
- Video format: MOV

It uses a controlled vocabulary of 5 emergency words:
- YES, NO, PAIN, WATER, HELP

Datasets








```
GT     : set green by h four again
Greedy : set green by h four again
Beam   : set green by h four again
--------------------------------------
GT     : set green by n five now
Greedy : set green by a five now
Beam   : set green by a five now
--------------------------------------
GT     : set green by n six soon
Greedy : set green by a six soon
Beam   : set green by a six soon
--------------------------------------
GT     : set green by n seven please
Greedy : set green by s seven please
Beam   : set green by s seven please
--------------------------------------
GT     : set green by n eight again
Greedy : set green by a eight again
Beam   : set green by a eight again
--------------------------------------
GT     : set green by t nine now
Greedy : set green by t nine now
Beam   : set green by t nine now
--------------------------------------
GT     : set green by u one please
Greedy : set green by j one please
Beam   : set green by j one please
```

- **Greedy**: Picks the most likely character at each step (fast, local decision).
- **Beam**: Evaluates multiple candidate sequences to find the most probable sentence (more accurate).

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Model Evaluation

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GRID Dataset

- **Evaluated Samples: 300**
- **Character Accuracy: 93.93%**

Custom Dataset

- **Evaluated Samples: 110**
- **Test Accuracy : 87.61%**

Work Division

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Module	Description	Assigned To
Input Capturing	Capture real-time video focusing on mouth region using camera. Use MediaPipe for ROI localization.	Giribala (Frontend + Training Support)
Preprocessing	Denoise frames, adjust brightness, normalize contrast. Convert frames to grayscale.	Darsan (Backend + Model Training)
Visual Speech Recognition	Develop deep learning models. Predict phonemes/words/sentences from silent speech.	Geevar (Backend + Model Training)
Ouput	Implement real-time transcription display. Handle continuous live updates.	Fathima Meharin (Frontend + Training Support)

	July	August	September	October	November	December	January	February
Planning	Collective							
Literature Review	Collective							
Pre-Trained Models	Darsan & Geevar							
Data Collection		Collective						
Front-end Setup		Giribala & Meharin						
30% Completion			Collective					
50% Completion					Collective			
Paper & Patent					Giribala & Meharin			
75% Completion						Darsan & Geevar		
Real-Time Setup							Collective	

Gantt Chart

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Technical Requirements

Hardware:

- User: High-Resolution Cameras (standard camera).
- Developer: 8GB RAM, RTX 1650.
- Smartphones | Tablets | Laptops.

Software:

- Programming Language: Python (Python 3.10.18).
- Machine Learning Frameworks: PyTorch (2.5.1+cu121).
- Facial Landmark Detection: MediaPipe (0.10.21).
- Text-to-Speech Engines: pyttsx3 (Offline Text-to-Speech Library).
- Environment Manager: Anaconda (conda 25.7.0).

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Risk and Challenges

- **Lighting Issues:** Poor lighting can reduce the accuracy of mouth detection and recognition.
- **Variability in Lip Movements:** Different users articulate speech differently, affecting model performance.
- **Head Movements:** Excessive movement can cause tracking errors during video capture.
- **Limited Datasets:** Lack of large, annotated lip-reading datasets can hinder model training.
- **High Computational Demand:** Deep learning models require powerful hardware for real-time processing.

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Expected Output

- **Recognized Text Display:** The system should accurately convert the user's lip movements into readable text.
- **Real-Time Output:** The recognized text should be displayed either live during video capture or immediately after processing.

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Conclusion

A real-time lip-reading application that translates lip movements into readable and audible communication. Enables non-verbal communication through live video. Offers both text display and optional speech conversion. Runs on common devices without the need for invasive tech.

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

PO2: **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.

PO4: **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, while recognizing their limitations to solve complex engineering problems.

PO6: **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: **Individual and Collaborative Team Work:** Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.

PO9: **Communication:** Communicate effectively and inclusively within the engineering community and society at large, including the ability to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.

PO10: **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: **Life-Long Learning:** Recognize the need for and have the preparation and ability for (i) independent and lifelong learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering program will demonstrate:

PSO1: **Computer Science Specific Skills**
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: **Programming and Software Development Skills**
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: **Professional Skills**
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet societal needs, thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

CO1: Model and solve real world problems by applying knowledge across domains
(Cognitive knowledge level: Apply).

CO2: Develop products, processes or technologies for sustainable and socially relevant applications
(Cognitive knowledge level: Apply).

CO3: Function effectively as an individual and as a leader in diverse teams and comprehend and execute designated tasks
(Cognitive knowledge level: Apply).

CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms
(Cognitive knowledge level: Apply).

CO5: Identify technology or research gaps and propose innovative or creative solutions
(Cognitive knowledge level: Analyze).

CO6: Organize and communicate technical and scientific findings effectively in written and oral forms
(Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

Table 6.1: Course Outcomes

S.No	Description	Bloom's Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative or creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

CO–PO Mapping

Table 6.2: CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	1	2	1	1	1	1	1	2
CO2	2	2	2	2	3	2	1	1	1	1	2
CO3								3	2	2	1
CO4		1	2		3	2	3	2	2	3	1
CO5	2	3	3	1	2					1	3
CO6			2			2	2	2	3	1	1

CO–PSO Mapping

Table 6.3: CO–PSO Mapping

CO	PSO1	PSO2	PSO3
CO1	2	2	2
CO2	2	2	
CO3			3
CO4		2	2
CO5			
CO6			3

Justifications for CO–PO Mapping

Mapping	Level	Justification
101003/CS822U CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/CS822U CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/CS822U CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/CS822U CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/CS822U CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/CS822U CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/CS822U CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/CS822U CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/CS822U CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/CS822U CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/CS822U CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/CS822U CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/CS822U CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/CS822U CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/CS822U CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/CS822U CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.

Mapping	Level	Justification
101003/CS822U CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/CS822U CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/CS822U CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/CS822U CO2-PO8	Low	Development activities may require collaboration among team members.
101003/CS822U CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/CS822U CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/CS822U CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/CS822U CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/CS822U CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/CS822U CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/CS822U CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/CS822U CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/CS822U CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/CS822U CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/CS822U CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/CS822U CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/CS822U CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/CS822U CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/CS822U CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.

Mapping	Level	Justification
101003/CS822U CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/CS822U CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/CS822U CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/CS822U CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/CS822U CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/CS822U CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/CS822U CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/CS822U CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/CS822U CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/CS822U CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/CS822U CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/CS822U CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/CS822U CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/CS822U CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/CS822U CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/CS822U CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/CS822U CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/CS822U CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/CS822U CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.

Mapping	Level	Justification
101003/CS822U CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/CS822U CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/CS822U CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.